A Supplements

In this supplement, we provide additional theoretical insights and technical details to complement the main paper. We first discuss how our theoretical guarantees position VANEB relative to recent Bayesian approaches in personalized federated learning and foundational works in distributed statistical inference. We then present a detailed derivation of the generalized Tweedie formula that underpins the VANEB estimator, along with a comprehensive description of the algorithmic implementation and its convergence properties. Finally, we outline how we adapt the VANEB framework for neural network training, including practical considerations for estimating variance in this context.

A.1 Comparisons to Related Work

Our theoretical guarantees highlight the distinct advantages of VANEB when compared to recent parametric Bayesian approaches in Personalized Federated Learning (PFL) and foundational works in distributed statistical inference.

Bayesian Personalized Learning. Recent Bayesian PFL methods, most notably Ozkara et al. [2023], Kotelevskii et al. [2022, FedPop] and Wu et al. [2023, FedGMM], share our motivation of framing personalization as a hierarchical Bayesian inference problem. Ozkara et al. [2023] establishes a robust theoretical framework by assuming a homoskedastic Gaussian observation model ($\Sigma_k \equiv \sigma_x^2 \mathbf{I}_d$) alongside a known, single-component Gaussian prior $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \sigma_\theta^2 \mathbf{I}_d)$. Under these strict parametric assumptions, the personalized estimator admits a closed-form convex combination, achieving an exact Bayes Mean Squared Error (MSE) bounded by $\mathcal{O}(1/n)$. While their practical algorithms explore mixture distributions to improve empirical flexibility, their exact theoretical guarantees remain strictly confined to this single-Gaussian regime, which fundamentally underfits and fails to capture the underlying structural nuances.

Kotelevskii et al. [2022] advocates for more flexible parametric energy-based priors, including Gaussian Mixture Models (GMMs), utilizing MCMC or variational inference. Yet, its prior is constrained to have an exponentially decaying tail with respect to energy, and it primarily assumes parameter-independent (homoskedastic) noise. Similarly, Wu et al. [2023] employs a finite Gaussian Mixture Model (GMM). While more flexible, their theoretical guarantees heavily rely on correctly specifying the true number of mixture components and imposing strict parametric assumptions on the base distributions. Consequently, they remain susceptible to misspecification bias if the parameters lie on continuous, non-linear low-dimensional manifolds (Figure 1). In contrast, the VANEB oracle inequality in (14) makes no parametric assumptions on the latent prior G_0. By learning the prior nonparametrically via the NPMLE, VANEB autonomously adapts to a broad class of underlying prior geometries—avoiding the need to pre-specify the number of mixture components. We show that the statistical price for this flexibility is a poly-logarithmic penalty in K, achieving an asymptotic denoising rate of $\tilde{\mathcal{O}}(\frac{1}{n}\varepsilon_M^2(S_\bullet, G_0))$ that preserves the collaborative pooling benefit across K clients, while robustly accommodating parameter-dependent heteroskedasticity.

Distributed Statistical Inference. Beyond PFL, recent advances in distributed statistical inference have rigorously addressed the challenges of data heterogeneity from a classical statistical perspective. For instance, Gu and Chen [2023, 2024] establish the asymptotic normality and statistical efficiency of distributed M-estimators and decentralized federated learning algorithms. Their foundational work elegantly corrects for heterogeneity-induced biases to achieve optimal inference for a single, global consensus parameter. Meanwhile, our VANEB framework pivots toward *personalized* inference under heterogeneity. Instead of applying bias corrections to force local estimates to align with a single global model, VANEB models this structural heterogeneity by formulating the local M-estimators $\hat{\boldsymbol{\theta}}_k$ as noisy observations drawn from an unknown, complex prior G_0. Consequently, while Gu and Chen [2023, 2024] provide optimal inferential guarantees for the global aggregate, our theoretical results establish oracle denoising bounds for the individualized client parameters.

A.2 Generalized Tweedie Formula and VANEB Estimator

Our aim is to obtain personalized estimators $\hat{\boldsymbol{\theta}}_1, ..., \hat{\boldsymbol{\theta}}_K$ that minimize the Bayes risk:

$$\mathcal{R}_{G_0}(\hat{\boldsymbol{\theta}}) := \frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\|\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k\|^2$$

and naturally this reduces to providing Bayes rule $\mathbb{E}[\boldsymbol{\theta}_k \mid \underline{\hat{\boldsymbol{\theta}}}_k]$ for each client. Under our Gaussian likelihood setup, the posterior mean is given by the Tweedie formula [Efron, 2014]. When the covariance is a fixed matrix Σ_k, the standard identity $\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) = \Sigma_k \nabla f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)$ holds (since Σ_k

can be pulled outside the integral), and one recovers the formula of Soloff et al. [2025]:

$$\mathbb{E}(\boldsymbol{\theta}_k|\hat{\underline{\boldsymbol{\theta}}}_k) = \frac{\hat{\underline{\boldsymbol{\theta}}}_k}{\sqrt{n_k}} + \frac{\Sigma_k}{\sqrt{n_k}} \nabla \log f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k).$$

However, when $\Sigma_k = \Sigma_k(\boldsymbol{\theta})$ depends on the parameter, this identity no longer holds. Our generalized heteroskedastic extension is essential for this setting. Therefore, we propose the VANEB estimator to be a generalized Tweedie formula that incorporates the parameter-dependent covariance structure. The key is to express the scaled score vector $\mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)$ and the marginal density $f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)$ as functionals of the likelihood integrated against the prior distribution G_0:

$$\mathbf{q}_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k) := \int \Sigma_k(\boldsymbol{\theta}) \frac{\partial}{\partial \hat{\underline{\boldsymbol{\theta}}}_k} \varphi^{(k)}(\hat{\underline{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})\, G_0(\mathrm{d}\boldsymbol{\theta}),\ \ f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k) = \int \varphi^{(k)}(\hat{\underline{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})\, G_0(\mathrm{d}\boldsymbol{\theta})$$

By the diagonal structure, we can write

$$\mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k) = \int \begin{bmatrix} \sigma_{k,1}(\boldsymbol{\theta})^2 \frac{\partial}{\partial \hat{\underline{\theta}}_1} \\ \vdots \\ \sigma_{k,d}(\boldsymbol{\theta})^2 \frac{\partial}{\partial \hat{\underline{\theta}}_d} \end{bmatrix} \varphi^{(k)}(\hat{\underline{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) G(\mathrm{d}\boldsymbol{\theta}) =: \begin{bmatrix} \frac{\partial}{\partial \hat{\underline{\theta}}_1} \tilde{f}_{k,G,1}(\hat{\underline{\boldsymbol{\theta}}}_k) \\ \vdots \\ \frac{\partial}{\partial \hat{\underline{\theta}}_d} \tilde{f}_{k,G,d}(\hat{\underline{\boldsymbol{\theta}}}_k) \end{bmatrix}$$

where $\tilde{f}_{k,G,i}(\hat{\underline{\boldsymbol{\theta}}}_k) := \int \sigma_{k,i}(\boldsymbol{\theta})^2 \varphi^{(k)}(\hat{\underline{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) G(\mathrm{d}\boldsymbol{\theta})$ represents the *variance-weighted marginal density* for coordinate i. The factor $\sigma_{k,i}(\cdot)^2$ accounts for the parameter-dependent variance in coordinate i; in homoskedastic settings, this reduces to a constant, and the formula collapses to the classical Tweedie formula. This decomposition allows the scaled score to be expressed as a gradient of component-specific functions, which is essential for handling the heteroskedastic variance structure.

By direct computation, we expand the expected value of the scaled score:

$$\mathbf{q}_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k) = \int \varphi^{(k)}(\hat{\underline{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \cdot (\sqrt{n_k}\boldsymbol{\theta} - \hat{\underline{\boldsymbol{\theta}}}_k)\, G_0(\mathrm{d}\boldsymbol{\theta}).$$

From this decomposition, we obtain a generalization of Tweedie's formula that accommodates heteroskedastic, parameter-dependent noise:

$$\sqrt{n_k}\mathbb{E}(\boldsymbol{\theta}_k|\hat{\underline{\boldsymbol{\theta}}}_k) = \hat{\underline{\boldsymbol{\theta}}}_k + \frac{\mathbf{q}_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)},$$

which in the original scale becomes

$$\boldsymbol{\theta}_k^o := \mathbb{E}(\boldsymbol{\theta}_k | \underline{\hat{\boldsymbol{\theta}}}_k) = \frac{\hat{\boldsymbol{\theta}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{\sqrt{n_k} f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}.$$

$\boldsymbol{\theta}_k^o$ is the oracle Bayes estimator for client k under the true prior G_0. Compared with the classical Tweedie formula, the correction term here is no longer a fixed-variance score; it incorporates the local uncertainty through the coordinate-wise weights $\sigma_{k,i}(\boldsymbol{\theta})^2$ through $\tilde{f}_{k,G,i}$. It is precisely this weighting that makes the shrinkage adaptive to the local noise level of each client.

The oracle Bayes estimator $\boldsymbol{\theta}_k^o$ depends on the unknown prior G_0, so the server must estimate G_0 from the estimates $\{\underline{\hat{\boldsymbol{\theta}}}_k\}_{k=1}^K$ and covariance functions $\{\Sigma_k(\cdot)\}_{k=1}^K$. We do this with an NPMLE. Given an estimated prior $\hat{G}$, the resulting VANEB personalized estimator becomes

$$\hat{\boldsymbol{\theta}}_k^v := \frac{\hat{\boldsymbol{\theta}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{\sqrt{n_k} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}$$

where the NPMLE $\hat{G}$ serves as a data-driven estimator for the unknown prior distribution G_0. We develop the NPMLE for our heteroskedastic setting and establish its convergence properties.

Our generalized formula replaces the fixed-covariance score with a $\Sigma_k(\cdot)$-weighted average score that accounts for parameter-dependent uncertainty. In a high-variance coordinate (large $\sigma_{k,i}$), the gradient is small so the posterior is dominated by the prior: the update is *large*, pulling the estimate strongly toward the global center. In a low-variance coordinate (small $\sigma_{k,i}$), the likelihood is sharp so the posterior stays near the local estimate: the update is *small*. This "borrow more where the local estimate is unreliable" principle—a cornerstone of James–Stein estimation and empirical Bayes shrinkage [James and Stein, 1961, Stein, 1956, Efron, 2012]—allows VANEB to effectively balance personalization and information pooling, leading to improved cohort performance.

Compared with common FL optimizers, this update has a distinct role. FedAvg [McMahan et al., 2017a] uses isotropic averaging of local steps, and FedProx/SCAFFOLD [Li et al., 2020, Karimireddy et al., 2020] stabilize optimization drift through proximal regularization or control variates; all are primarily designed around iterative gradient optimization. In contrast, VANEB applies a one-shot empirical Bayes correction to the local estimates $\{\hat{\boldsymbol{\theta}}_k\}$ (which may be MLEs, M-estimators, or any asymptotically normal estimator; Van der Vaart [2000]), where the direction is modulated by client-specific uncertainty geometry. Relative to personalized SGD-style methods (e.g., pFedMe/Per-FedAvg [T Dinh et al., 2020, Fallah et al., 2020]), VANEB therefore shifts the main burden from learning-rate tuning and repeated local iterations to estimating the shared prior G_0. This makes it particularly appealing in settings where communication is expensive or local computation is constrained, as it requires only a single round of communication for the local estimates and a single global update for personalization, without the need for multiple rounds of local training or fine-tuning.

A.3 Computation of VANEB Estimator

In the classical setting where Σ_k is independent of $\boldsymbol{\theta}$, the ridgeline manifold $\mathcal{M}$ of Lemma 3.2 characterizes the NPMLE atoms exactly: the score equation is linear in the atom and the M-step reduces to a closed-form weighted average.

However, when $\Sigma_k(\boldsymbol{\theta})$ genuinely depends on $\boldsymbol{\theta}$, the score of the mixture density $\psi(\boldsymbol{\theta}) = \sum_{k=1}^{K} w_k \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})$ is the weighted sum of

$$\nabla\psi_k(\boldsymbol{\theta}) := \sqrt{n_k}\Sigma_k(\boldsymbol{\theta})^{-1}(\underline{\hat{\boldsymbol{\theta}}}_k - \sqrt{n_k}\boldsymbol{\theta}) + \sum_{i=1}^{d} \frac{1}{\sigma_{k,i}(\boldsymbol{\theta})} \left\{ \frac{(\underline{\hat{\theta}}_{k,i} - \sqrt{n_k}\theta_i)^2}{\sigma_{k,i}(\boldsymbol{\theta})^2} - 1 \right\} \nabla\sigma_{k,i}(\boldsymbol{\theta}),$$

so unlike Ray and Lindsay [2005], Saha and Guntuboyina [2020], Soloff et al. [2025], the equation $\sum_{k=1}^{K} w_k \nabla\psi_k(\boldsymbol{\theta}) = \mathbf{0}_d$ involves additional gradients with respect to variance functions $\sigma_{k,i}$'s, and no closed-form solution exists for the true critical locus.

Algorithm 2 addresses this intractability by alternating between (i) solving a fixed-covariance NPMLE subproblem—where substituting $\mathbf{S}_k = \Sigma_k(\mathbf{a}_j)$ recovers the tractable ridgeline property—and (ii) dynamically updating the covariances evaluated at the newly estimated atom locations. This iterative scheme produces, at convergence, a fixed point of the coupled atom-covariance system and can be viewed as an approximation of the intractable heteroskedastic locus by a sequence of computable ridgeline manifolds.

A.3.1 Algorithm

A practical implementation must therefore keep the atom locations and their client-specific covariances synchronized throughout the pseudo-EM loop. Standard NPEB software cannot do this directly because it assumes precisions are fixed once the support grid is chosen. Our implementation instead uses a fixed-covariance conic solve only as an initialization, followed by a custom pseudo-EM routine in which

Algorithm 2: VANEB Estimator

Input: Observations $\{\hat{\boldsymbol{\theta}}_k\}_{k=1}^K$; covariance function Σ_k; Pseudo-EM iterations T

Output: Personalized estimates $\{\hat{\boldsymbol{\theta}}_k^v\}_{k=1}^K$

Initialize atoms $\{\mathbf{a}_j\}_{j=1}^m \leftarrow \{\hat{\boldsymbol{\theta}}_k\}_{k=1}^K$ and covariance $\Sigma_{k,j} \leftarrow \Sigma_k(\mathbf{a}_j)$

Fit NPMLE weights: $\hat{\mathbf{w}} \leftarrow \arg\max_{\mathbf{w}} \sum_{k=1}^K \log \sum_{j=1}^m w_j\, \varphi^{(k)}(\sqrt{n_k}\hat{\boldsymbol{\theta}}_k; \mathbf{a}_j)$ **for** $j = 1, \ldots, m$ **do**

  for $t = 1, \ldots, T$ **do**

    E-step: $r_{kj} \propto \hat{w}_j\, \varphi^{(k)}(\sqrt{n_k}\hat{\boldsymbol{\theta}}_k; \mathbf{a}_j)$

    M-step: $\mathbf{a}_j \leftarrow \dfrac{\sum_{k=1}^K r_{kj}\, [\Sigma_{k,j}/n_k]^{-1} \hat{\boldsymbol{\theta}}_k}{\sum_{k=1}^K r_{kj}\, [\Sigma_{k,j}/n_k]^{-1}}$

    Recompute covariances: $\Sigma_{k,j} \leftarrow \Sigma_k(\mathbf{a}_j)$ `// Atom-covariance coupling`

    Update weights: $\hat{w}_j \leftarrow \dfrac{1}{K} \sum_{k=1}^K r_{kj}$

return $\hat{\boldsymbol{\theta}}_k^v \leftarrow \sum_{j=1}^m r_{kj}\, \mathbf{a}_j$ for all k

1. the E-step recomputes responsibilities using the current atom-dependent likelihoods,

2. the M-step updates each atom by precision-weighted averaging, and

3. the covariance field is re-evaluated at the updated atoms before the next sweep.

The resulting posterior means are exactly the heteroskedastic empirical Bayes estimates under the fitted discrete prior. Algorithm 2 summarizes this procedure.

The final estimate produced by Algorithm 2,

$$\hat{\boldsymbol{\theta}}_k^v = \sum_{j=1}^m r_{kj}\, \mathbf{a}_j = \frac{\sum_{j=1}^m \hat{w}_j\, \mathbf{a}_j\, \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{a}_j)}{\sum_{j=1}^m \hat{w}_j\, \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{a}_j)},$$

is precisely the posterior mean under the fitted discrete mixture prior $\hat{G} = \sum_{j=1}^m \hat{w}_j\, \delta_{\mathbf{a}_j}$ and the heteroskedastic likelihood $\varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})$. This is mathematically equivalent to applying the modified heteroskedastic Tweedie formula with the NPMLE $\hat{G}$ in place of G_0:

$$\hat{\boldsymbol{\theta}}_k^v = \frac{\int \boldsymbol{\theta}\, \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})\, \mathrm{d}\hat{G}(\boldsymbol{\theta})}{\int \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})\, \mathrm{d}\hat{G}(\boldsymbol{\theta})} = \frac{\underline{\hat{\boldsymbol{\theta}}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{\sqrt{n_k} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}.$$

The Pseudo-EM iterations refine the atom locations and weights data-adaptively, while the Tweedie representation provides a closed-form expression for the resulting posterior mean, which facilitates the theoretical understanding. Both perspectives yield the same denoised estimate; the algorithm offers a practical route to computing it without explicitly evaluating the ratio of marginal densities.

A.3.2 Computation

Following Koenker and Mizera [2014], we restrict the NPMLE to a support-constrained version. For any non-empty, closed set $\mathcal{A} \subseteq \mathbb{R}^d$, define the support-constrained NPMLE as

$$\hat{G}^{\mathcal{A}} \in \arg\max_{G \in \mathcal{P}(\mathcal{A})} \ell_K(G), \tag{15}$$

where $\mathcal{P}(\mathcal{A})$ denotes the collection of probability measures supported on $\mathcal{A}$. By construction, $\hat{G} = \hat{G}^{\mathbb{R}^d}$.

For computation, the key tuning parameter is the grid resolution δ. As in Soloff et al. [2025], we discretize the ridgeline region $\mathcal{M}$ by covering it with axis-aligned hypercubes:

$$\mathcal{H} = \left\{ \mathbf{H}_j + [-\delta/2, \delta/2]^d : j \in \{1, \dots, J\} \right\},$$

where the centers $\{\mathbf{H}_1, \dots, \mathbf{H}_J\}$ are spaced δ apart. The candidate support $\mathcal{A}$ is then the set of all 2^d corner points of each hypercube, i.e., $\mathbf{H}_j + \frac{\delta}{2}\mathbf{v}$ for $\mathbf{v} \in \{-1, 1\}^d$. Since $\mathcal{M}$ is compact, $\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ is finite, and the support-constrained NPMLE reduces to a finite-dimensional optimization over mixing weights: $\hat{G}^{\mathcal{A}} = \sum_{j=1}^m \hat{w}_j \delta_{\mathbf{a}_j}$ where

$$\hat{\mathbf{w}} \in \arg\max_{\mathbf{w} \in \Delta_{m-1}} \frac{1}{K} \sum_{k=1}^{K} \log \left(\sum_{j=1}^{m} w_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{a}_j) \right). \tag{16}$$

In Soloff et al. [2025], the precision matrices stay fixed throughout the optimization. Here they must move with the atoms. The role of the discretized solve is therefore only to provide a stable initilization for the support and weights; the subsequent pseudo-EM iterations in Algorithm 2 performs the actual heteroskedastic recoupling.

The following proposition quantifies the approximation error introduced by this discretization. As $\delta \to 0$, the discretized NPMLE's log-likelihood converges to that of the unrestricted NPMLE, with an explicit bound that can guide practical grid selection:

Proposition A.1 (Soloff et al. [2025]). *Let $\mathcal{M} \subset \mathbb{R}^d$ be any compact set containing all solutions to the true NPMLE problem (2). Assume the diameter of $\mathcal{M}$ is at most D and fix grid spacing $\delta \in \left(0, \sqrt{\frac{3}{4d}\frac{s}{D}}\right)$. Construct a grid covering $\mathcal{H}$ of $\mathcal{M}$ with hypercubes of width δ, and let $\mathcal{A}$ be the set of corner points. Then any discretized NPMLE solution $\hat{G}^{\mathcal{A}}$ from (15) satisfies the* *approximation error bound*

$$\sup_{G\in\mathcal{P}(\mathbb{R}^d)} \ell_K(G) - \ell_K(\hat{G}^{\mathcal{A}}) \le \frac{d}{\underline{s}}\left(\frac{1}{2} + \frac{2D^2}{\underline{s}}\right)\delta^2. \tag{17}$$

A.3.3 Convergence of VANEB Iterates

We next state a fixed-point guarantee for the iterative scheme in Algorithm 2. The lemma has two parts: the first concerns existence and uniqueness of the frozen-covariance map under contraction, and the second quantifies how far the limiting atoms can be from true critical points of the full score.

Lemma A.1 (Convergence of VANEB iterates)**.** *Assume $\boldsymbol{\theta} \mapsto \Sigma_k(\boldsymbol{\theta})$ is L_Σ-Lipschitz uniformly in k with eigenvalues in $[\underline{s}, \bar{s}]$. Define the VANEB iteration: given atoms $\mathbf{a}^{(t)}$, set*

$$\mathbf{a}_j^{(t+1)} = \bar{\boldsymbol{\psi}}_j(\mathbf{a}^{(t)}) := \left(\sum_k w_{kj}^{(t)} n_k \Sigma_k(\mathbf{a}_j^{(t)})^{-1}\right)^{-1} \sum_k w_{kj}^{(t)} n_k \Sigma_k(\mathbf{a}_j^{(t)})^{-1} \hat{\boldsymbol{\theta}}_k,$$

where $w_{kj}^{(t)} \propto w_k \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{a}_j^{(t)})$, $\sum_{k=1}^K w_{kj}^{(t)} = 1$. Write $R := \max_{k,\ell}\|\hat{\boldsymbol{\theta}}_k - \hat{\boldsymbol{\theta}}_\ell\|$ and $L_{\bar{\psi}} := \bar{s}RL_\Sigma/\underline{s}^2$. Then, the following holds.

1. *Each atom satisfies*
$$\sum_{k=1}^K \bar{w}_{kj}^{(t)} n_k \Sigma_k(\mathbf{a}_j^*)^{-1}(\hat{\boldsymbol{\theta}}_k - \mathbf{a}_j^*) = \mathbf{0}_d$$
for some weights $\bar{\mathbf{w}}_j^{(t)} = \{\bar{w}_{kj}^{(t)}\}_{k=1}^K$ with $\bar{w}_{kj}^{(t)} \ge 0$ and $\sum_{k=1}^K \bar{w}_{kj}^{(t)} = 1$. If $L_{\bar{\psi}} < 1$, the fixed point is unique with linear rate $L_{\bar{\psi}}^t$.

2. *Assume, in addition, that Σ_k is continuously differentiable. Then for each atom j there exists a true critical point $\mathbf{a}_{j0}^*$ of the full heteroskedastic score, $\sum_{k=1}^K w_{kj}^{(t)} \nabla\psi_k(\mathbf{a}_{j0}^*) = \mathbf{0}_d$, with*
$$\|\mathbf{a}_{j0}^* - \mathbf{a}_j^*\| = \mathcal{O}_P\left(\frac{d^{3/2}\bar{s}L_\Sigma}{\underline{s}\, n_j^w}\right) = o_P(1) \quad \textit{as} \quad \underline{n} \to \infty, \tag{18}$$
where $n_j^w := \sum_{k=1}^K w_{kj}^{(t)} n_k$ is the effective sample size at atom j.

The first claim shows that the frozen-covariance map $\bar{\boldsymbol{\psi}}$ inherits the self-mapping property of the static ridgeline, so that Algorithm 2 always produces a well-defined limit—without requiring the contraction condition $L_{\bar{\psi}} < 1$, which may fail when clients are heterogeneous (R large) or covariances vary steeply (L_Σ large). The second claim quantifies the cost of freezing: the converged atoms $\mathbf{a}_j^*$ are not exact stationary points of the full heteroskedastic score, but the discrepancy $\|\mathbf{a}_{j0}^* - \mathbf{a}_j^*\|$ vanishes at rate $\mathcal{O}_P(1/n_j^w)$, i.e., at the parametric rate in the effective local sample size. Crucially, the self-regularizing of the posterior weights—which exponentially downweight clients far from each atom—ensures that n_j^w grows with the minimum sample size $\underline{n}$ and that the residual is controlled by the *local* signal-to-noise ratio rather than by the global diameter R or the number of clients K. In summary, VANEB retains the computational tractability of static-covariance NPEB (each iteration solves a standard ridgeline problem) while converging to the heteroskedastic solution as sample sizes grow.

B Neural Network Adaptations

In this section, we extend our Variance-Aware Nonparametric Empirical Bayes (VANEB) framework to federated training of deep neural networks. While the asymptotic normality assumption central to our VANEB theory does not strictly hold for highly non-convex neural network architectures, we heuristically apply our framework by treating the local weight estimates as approximately Gaussian. This allows us to investigate whether the VANEB penalty can act as an effective, data-driven regularizer even in spaces where the theoretical assumptions are violated.

We propose two algorithms, VANEB-full and VANEB-head. In the VANEB-full method, the empirical Bayes personalization procedure is applied to the entire weight vector of the neural network. Concretely, at each communication round, clients train local weights $\boldsymbol{\omega}_k$ for the neural network and communicate $\boldsymbol{\omega}_k$ and $\hat{\Sigma}_k$ to the server. The server computes the responsibilities and the atoms through Pseudo-EM iterations as described previously for M-estimators using the Gaussian density function. The initial number of atoms is chosen to be $m = \max(K, 100)$. Once the responsibilities r_{kj} and atoms $\mathbf{a}_j$ are calculated, the server calculates personalized estimates $\boldsymbol{\omega}_k^v$ and returns those to the clients. The server also updates the global model as a weighted average of these $\boldsymbol{\omega}_k^v$.

A difficulty of applying VANEB with deep neural networks is that we typically do not know the variance of the weight vectors, not even as a function $\Sigma(\boldsymbol{\omega})$ of $\boldsymbol{\omega}$. Therefore, we approximate the variance at the clients as,

$$\hat{\Sigma}_k = \mathrm{diag}\left[\left(\frac{1}{n_k}\sum_{i=1}^{n_k} \mathbf{g}_{k,i}\mathbf{g}_{k,i}{}^\top + \lambda \mathbf{I}\right)^{-1}\right],$$

where $\mathbf{g}_{k,i}$ is the gradient vector for the ith data point of client k. Note we only keep the diagonal elements and form a diagonal covariance matrix for efficiency in communication cost. A similar construction was recently proposed in Jhunjhunwala et al. [2024]. However, our procedure requires the variance as a (possibly client-dependent) function of $\boldsymbol{\omega}$, i.e., $\Sigma_k(\boldsymbol{\omega})$, so that it can be evaluated at the atoms $\Sigma_k(\mathbf{a}_j)$. We may either set $\Sigma_k(\mathbf{a}_j) = \hat{\Sigma}_k$ uniformly for any $\mathbf{a}_j$, or we can learn a function through a low-order polynomial regression of $\hat{\Sigma}_k$ against $\boldsymbol{\omega}_k$. The full algorithm is described in Algorithm 3.

Algorithm 3: VANEB-Full: Empirical Bayes Personalization with Full Network Weights

Input: Clients $k = 1, \ldots, K$; rounds T; initial model $\hat{\boldsymbol{\theta}}^{(0)}$; atoms m; local epochs E; damping λ

Output: Personalized models $\{\hat{\boldsymbol{\theta}}_k^{v,(t)}\}$

for $t = 0, \ldots, T-1$ **do**
- Server broadcasts $\hat{\boldsymbol{\theta}}^{(t)}$;
- **foreach** *client* k ***in parallel*** **do**
  - Train locally for E epochs $\Rightarrow \hat{\boldsymbol{\theta}}_k$;
  - Compute $\hat{\mathbf{F}}_k = \frac{1}{n_k}\sum_{i=1}^{n_k} \mathbf{g}_{ki}\mathbf{g}_{ki}^{\top}$ using local gradients $\mathbf{g}_{ki}$;
  - Set $\hat{\Sigma}_k = \operatorname{diag}\left((\hat{\mathbf{F}}_k + \lambda \mathbf{I})^{-1}\right)$;
  - Send $(\hat{\boldsymbol{\theta}}_k, \hat{\Sigma}_k, n_k)$ to server;
- Initialize atoms $\{\mathbf{a}_j\}_{j=1}^m$ and weights $\pi_j = 1/m$;
- Set $\Sigma_k(\mathbf{a}_j) \leftarrow \hat{\Sigma}_k$ for all k, j, or learn function through polynomial regression;
- **for** $s = 0, \ldots, S-1$ **do**
  - `// E-step`
  - **foreach** k, j **do**
    - $r_{kj} \propto \pi_j\, \varphi(\hat{\boldsymbol{\theta}}_k; \mathbf{a}_j, \Sigma_k(\mathbf{a}_j)/n_k)$;
  - Normalize $\{r_{kj}\}_j$;
  - `// M-step`
  - **for** $j = 1, \ldots, m$ **do**
    - $\pi_j \leftarrow \frac{1}{K}\sum_{k=1}^{K} r_{kj}$;
    - $\mathbf{a}_j \leftarrow \left(\sum_{k=1}^{K} r_{kj} n_k \Sigma_k(\mathbf{a}_j)^{-1}\right)^{-1} \sum_{k=1}^{K} r_{kj} n_k \Sigma_k(\mathbf{a}_j)^{-1}\hat{\boldsymbol{\theta}}_k$;
  - Update $\Sigma_k(\mathbf{a}_j)$ for all k, j;
- **foreach** *client* k **do**
  - $\hat{\boldsymbol{\theta}}_k^v \leftarrow \sum_{j=1}^{m} r_{kj}\mathbf{a}_j$;
- $\hat{\boldsymbol{\theta}}^{(t+1)} \leftarrow \sum_{k=1}^{K} \frac{n_k}{\sum_\ell n_\ell}\hat{\boldsymbol{\theta}}_k^v$;

return $\{\hat{\boldsymbol{\theta}}_k^{v,(t)}\}$;

However, updating the entire weight vector in the NPEB framework can be noisy, computationally expensive and communication inefficient. Hence we propose a second procedure which is a novel VANEB-head method that builds on the idea of separating the deep neural network as a shared backbone responsible for representation and a personalized last layer or head [Bengio, 2012, Tripuraneni et al., 2020, Collins et al., 2021]. The model is decomposed in a shared backbone $\boldsymbol{\psi}$ and a client specific last layer or head $\boldsymbol{\phi}_k$. Only the personalized parameters of the last layer are updated through the Empirical Bayes framework. The shared representation layers are updated as in FedAvg [McMahan et al., 2017a]. The full algorithm for the updating scheme of VANEB-head is described in Algorithm 4. We empirically observe the strong performance of VANEB head on various benchmark computer vision and language model fine-tuning datasets.

Algorithm 4: VANEB-Adapter: Empirical Bayes Personalization for Last Layer and LoRA updates

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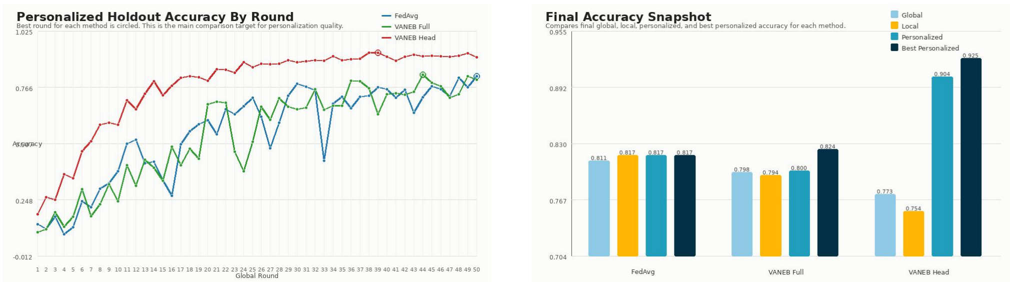
Input: Clients $k = 1, \dots, K$; rounds T; shared backbone $\boldsymbol{\psi}^{(0)}$; atoms m; local epochs E; damping λ

Output: Updated backbone $\boldsymbol{\psi}^{(T)}$ and personalized parameters $\{\tilde{\boldsymbol{\phi}}^{(k,t)}\}$

for $t = 0, \dots, T-1$ **do**
- Server broadcasts $\boldsymbol{\psi}^{(t)}$;
- **foreach** *client* k ***in parallel*** **do**
  - Train local model with shared backbone $\boldsymbol{\psi}^{(t)}$ and personalized parameter $\boldsymbol{\phi}_k$ for E epochs $\Rightarrow \hat{\boldsymbol{\phi}}_k, \Delta\boldsymbol{\psi}_k$;
  - Compute $\hat{\mathbf{F}}_{k,\phi} = \frac{1}{n_k} \sum_i \nabla_\phi \ell(z_{ki}; \hat{\boldsymbol{\phi}}_k) \nabla_\phi \ell(z_{ki}; \hat{\boldsymbol{\phi}}_k)^\top$;
  - Set $\hat{\Sigma}_k = \operatorname{diag}\left((\hat{\mathbf{F}}_{k,\phi} + \lambda \mathbf{I})^{-1}\right)$;
  - Send $(\hat{\boldsymbol{\phi}}_k, \hat{\Sigma}_k, n_k, \Delta\boldsymbol{\psi}_k)$ to server;
- $\boldsymbol{\psi}^{(t+1)} \leftarrow \text{Aggregate}(\{\Delta\boldsymbol{\psi}_k\}_{k=1}^K)$;
- Initialize atoms $\{\mathbf{a}_j\}_{j=1}^m$ and weights $\pi_j = 1/m$;
- Set $\Sigma_k(\mathbf{a}_j) \leftarrow \hat{\Sigma}_k$ for all k, j;
- **for** $s = 0, \dots, S-1$ **do**
  - `// E-step`
  - **foreach** k, j **do**
    - $r_{kj} \propto \pi_j \, \varphi(\hat{\boldsymbol{\phi}}_k; \mathbf{a}_j, \Sigma_k(\mathbf{a}_j)/n_k)$;
  - Normalize $\{r_{kj}\}_j$;
  - `// M-step`
  - **for** $j = 1, \dots, m$ **do**
    - $\pi_j \leftarrow \frac{1}{K} \sum_{k=1}^K r_{kj}$;
    - $\mathbf{a}_j \leftarrow \left(\sum_{k=1}^K r_{kj} n_k \Sigma_k(\mathbf{a}_j)^{-1}\right)^{-1} \left(\sum_{k=1}^K r_{kj} n_k \Sigma_k(\mathbf{a}_j)^{-1} \hat{\boldsymbol{\phi}}_k\right)$;
  - Update $\Sigma_k(\mathbf{a}_j)$ for all k, j;
- **foreach** *client* k **do**
  - $\tilde{\boldsymbol{\phi}}_k \leftarrow \sum_{j=1}^m r_{kj} \mathbf{a}_j$;
  - Send $\tilde{\boldsymbol{\phi}}_k$ to client k;

return $\boldsymbol{\psi}^{(T)}$ and $\{\tilde{\boldsymbol{\phi}}^{(k,t)}\}$;

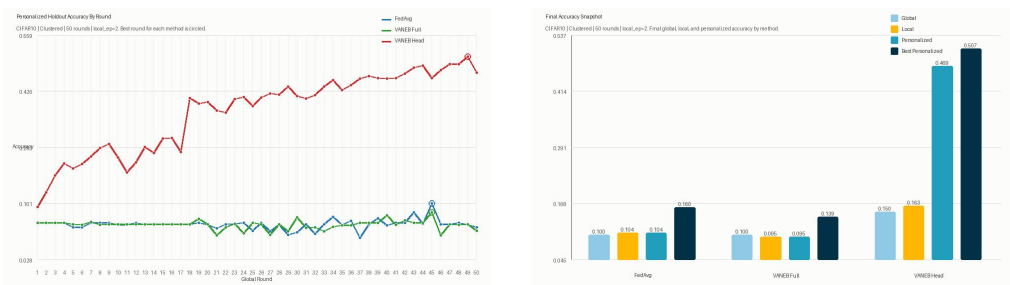
In both algorithms, the EB updates for personalized parameters are only done intermittently, while FedAvg updates are done in every global round of communication. In our data results, we



(a) Personalized holdout accuracy across global rounds for FedAvg, VANEB-Full, and VANEB-Head.

(b) Final and best accuracy summary across methods on the clustered MNIST setup.

Figure 4: Comparison of FedAvg, VANEB-Full, and VANEB-Head on clustered MNIST with 100 users, 10% participation per round, **2 local epochs**, and 250 samples per user.



(a) Personalized holdout accuracy across global rounds for FedAvg and VANEB-Head.

(b) Personalized and global model accuracy summary across methods.

Figure 5: Comparison of FedAvg, VANEB-Head, and local only methods on clustered CIFAR10 with 100 users, 10% participation per round, 2 local epochs, and 250 samples per user.

do an EB update for every 5 rounds of communications.

B.1 CNN on MNIST and CIFAR-10 vision datasets

We compare four methods on MNIST and CIFAR-10 using convolutional neural networks (CNNs): FedAvg, Local Only, VANEB-full, and VANEB-head. For MNIST, the model is a two-convolution CNN followed by two fully connected layers, with the final classification layer given by fully-connected layer 2 (`fc2`). For CIFAR-10, the model is a deeper CNN with three convolutional blocks followed by three fully connected layers, with the final classification layer given by the third fully-connected layer (`fc3`). In the VANEB-head method, this final linear layer is treated as the personalized head, while the remaining layers form the shared backbone.

We use a clustered heterogeneous client partition of the data. Clients are first divided into a small number of clusters, and each cluster is assigned its own pool of classes. Each client then receives data primarily from a small subset of classes drawn from its cluster-specific class pool, which induces structured label heterogeneity across clients. In addition, the training images are modified using cluster-dependent transformations to create feature heterogeneity across clusters. For MNIST, these transformations include rotations, translations, blur, and contrast changes. For CIFAR-10, they include color jitter, blur, and affine perturbations. After assigning examples to clients, each client's local dataset is split into a local training subset and a local holdout subset, and the local holdout data are used to evaluate personalization performance. We use 100 clients with each client receiving a maximum of 250 samples.

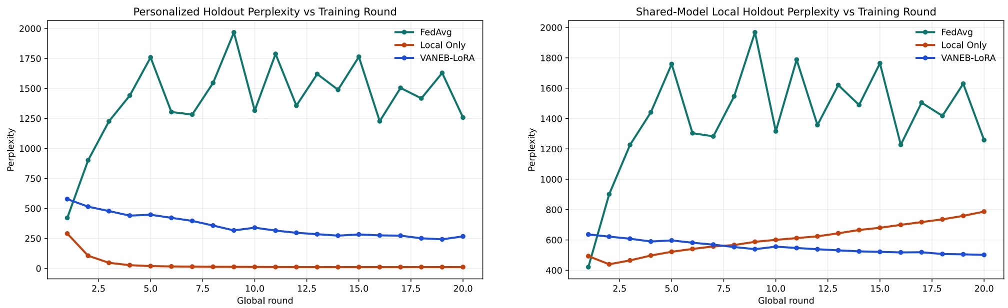
FedAvg trains local CNN models on sampled clients and aggregates all trainable parameters by weighted averaging. Local Only trains each client independently without cross-client aggregation. VANEB-full applies the empirical Bayes shrinkage step to the full trainable parameter vector of the CNN, while VANEB-head applies the empirical Bayes step only to the parameters of the final linear layer and updates the backbone by standard FedAvg. In both VANEB methods, empirical Bayes updates are performed intermittently, every `eb_every` communication rounds, while standard federated averaging steps are performed at every round.

We report three metrics during training: global test accuracy, obtained by evaluating the shared global model on the centralized test set; local-holdout accuracy, obtained by evaluating the shared global model on each client's holdout set; and personalized-holdout accuracy, obtained by evaluating each client's personalized model on its own holdout set. The personalized-holdout metric is the primary measure of personalization performance.

B.2 Fine-tuning a large language model with LoRA

We apply our VANEB approach to a parameter-efficient fine-tuning method for a large language model, called LoRA. For this we develop a method called VANEB-LoRA which adopts the LoRA method to the VANEB framework.

We evaluate federated causal language modeling on the Shakespeare dataset using DistilGPT2 [Sanh et al., 2019, Radford et al., 2019] with LoRA adaptation [Hu et al., 2022]. In this setup, each client corresponds to a Shakespeare character, and the data at that client consists of all text spoken by that character. We select the most active users up to the prescribed number of clients (100 for our results). Each example is formed by reconstructing the full text sequence from the provided prefix and next-character fields. The reconstructed text is then tokenized using the DistilGPT2 tokenizer and used for standard causal language model training. Thus,



(a) Personalized holdout accuracy across global rounds for FedAvg, VANEB-LoRA and local only methods.

(b) Perplexity in local holdout data using the global shared model.

Figure 6: Comparison of FedAvg, VANEB LoRA, and Local only methods for next token prediction task in terms of perplexity metric on Shakespeare data using a LoRA fine-tuning of 94M parameter DistillGPT2 model. The users are different characters in Shakespeare's novels

although the raw dataset is provided in prefix/next-character form, the implemented objective is next-token prediction over the full reconstructed sequence, and performance is evaluated using test perplexity. For each client, the assigned examples are split into a local training subset and a local holdout subset using an ordered split, so that the final portion of each client's sequence examples is reserved for holdout evaluation. We report the perplexity measure, with lower perplexity indicating better predictive performance.

$$\mathrm{PPL} = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log p(x_i \mid x_{<i})\right).$$

We consider three methods: FedAvg, Local Only, and VANEB-LoRA. In the pure-LoRA setting used here, the pretrained DistilGPT2 backbone is frozen, and only the LoRA adapter parameters are trainable.

Local Only achieved the best personalized holdout perplexity, reaching 10.59 at its best round and 10.86 at the final round, but at the cost of substantially worse shared-model behavior. VANEB-LoRA substantially improved over FedAvg on personalized perplexity, with final personalized holdout perplexity 266.75 versus 1258.04 for FedAvg, and also achieved the best final global test perplexity among the three methods, 366.95 versus 502.77 for Local Only and 768.28 for FedAvg. Thus, in this experiment, VANEB-LoRA provided a better balance between personalization and shared generalization than FedAvg and local-only training.

Algorithm 5: VANEB-LoRA for Federated LLM Fine-Tuning

Input: Frozen pretrained LM $\boldsymbol{\theta}_0$, client datasets $\{\mathcal{D}_k\}_{k=1}^{K}$, rounds T, participation fraction q, local epochs E, LoRA parameters $\mathbf{u}_k$, EB frequency b, number of atoms m, EM iterations L

Initialize: Initialize global frozen reference model with weights $\boldsymbol{\theta}_0$. Initialize client LoRA states $\mathbf{u}_k^{(0)}$ for all clients k.

for $t = 0, \ldots, T-1$ **do**
- Sample participating clients S_t with $|S_t| \approx qK$;
- **foreach** $k \in S_t$ **do**
  - Load client model $(\boldsymbol{\theta}_0, \mathbf{u}_k^{(t)})$;
  - Run local AdamW on $\mathcal{D}_k^{\text{train}}$ for E epochs, updating only LoRA parameters;
  - Obtain local LoRA vector $\hat{\boldsymbol{\theta}}_k^{(t)}$;
  - Estimate diagonal empirical Fisher $\hat{\mathbf{F}}_k^{(t)}$ on the LoRA coordinates using a small number of minibatches;
  - Send $(\hat{\boldsymbol{\theta}}_k^{(t)}, \hat{\mathbf{F}}_k^{(t)}, n_k)$ to the server;
- **if** $(t+1) \bmod b = 0$ **then**
  - Run VANEB shrinkage on $\{\hat{\boldsymbol{\theta}}_k^{(t)}, \hat{\mathbf{F}}_k^{(t)}, n_k : k \in S_t\}$;
  - Obtain personalized posterior means $\tilde{\mathbf{u}}_k^{(t+1)}$ for $k \in S_t$;
- **else**
  - Set $\tilde{\mathbf{u}}_k^{(t+1)} = \hat{\boldsymbol{\theta}}_k^{(t)}$ for $k \in S_t$;
- **foreach** $k \in S_t$ **do**
  - Update client state $\mathbf{u}_k^{(t+1)} \leftarrow \tilde{\mathbf{u}}_k^{(t+1)}$;
- **foreach** $k \notin S_t$ **do**
  - Keep previous client state $\mathbf{u}_k^{(t+1)} \leftarrow \mathbf{u}_k^{(t)}$;

C Proof of Main Results

C.1 Proof overview and comparison with prior work

Our proofs build on the general NPMLE analysis framework developed by Saha and Guntuboyina [2020] and Soloff et al. [2025] for the fixed-covariance (homoskedastic) setting. However, the extension to parameter-dependent covariances $\Sigma_k(\boldsymbol{\theta})$ introduces several structural difficulties that require new arguments:

- **Variance-weighted marginals.** The classical analysis works directly with $f_{k,G}$. In our setting, the oracle Bayes rule involves variance-weighted densities $\tilde{f}_{k,G,i}$ that have no counterpart in prior work. We introduce a sandwiching argument (Section C.3) to transfer Hellinger bounds between $f_{k,G}$ and $\tilde{f}_{k,G,i}$ via the condition number $\tau = \bar{s}/\underline{s}$.
- **Atom-covariance coupling.** The NPMLE support is no longer a tractable ridgeline manifold when Σ_k depends on $\boldsymbol{\theta}$. We establish convergence of the pseudo-EM iterates

under Lipschitz covariances (Lemma A.1) and quantify the approximation gap to the true heteroskedastic critical points via the implicit function theorem.

- **Lipschitz control of heteroskedastic kernels.** The metric entropy bounds require controlling the variation of $\varphi^{(k)}(\hat{\boldsymbol{\theta}};\boldsymbol{\theta})$ as a function of $\boldsymbol{\theta}$ when the covariance also moves. Lemma D.2 provides a new mean-value-type bound involving the Jacobian of the variance function, which replaces the standard Lipschitz bounds available in the fixed-covariance case.

- **Entropy inflation.** The covering numbers of the marginal class $\mathbb{F}$ grow polynomially in $\underline{n}^{d/2}$ due to the sharpening of kernels with increasing local sample size—a phenomenon absent when each client contributes a single observation as in Saha and Guntuboyina [2020], Soloff et al. [2025].

C.2 Notation

Let $\mathbb{B}_r(\hat{\boldsymbol{\theta}}) = \{\mathbf{y} \in \mathbb{R}^d : \|\hat{\boldsymbol{\theta}} - \mathbf{y}\| \leq r\}$ denote a closed ball in $\mathbb{R}^d$. For an integer $m > 0$, we use a generic notation $[m] = \{1, \ldots, m\}$. We adopt the following notation from Soloff et al. [2025]:

1. Given a pseudo-metric space (M, m) and $\varepsilon > 0$, let $N(\varepsilon, M, m)$ denote the ε-covering number:
$$N := \arg\min_N \left\{ M \subset \bigcup_{i=1}^{N} \{y : m(y, x_i) \leq \varepsilon\}; x_1, \ldots, x_N \in M \right\}.$$
When M is a subset of $\mathbb{R}^d$ we write $N(\varepsilon, M)$ instead of $N(\varepsilon, M, \|\cdot\|)$.

2. We use the shorthand $f_{\bullet,G} = (f_{k,G})_{k=1}^K$ and $\mathbf{q}_{\bullet,G} = (\mathbf{q}_{k,G})_{k=1}^K$ analogously. Also, define the spaces of marginals $f_{\bullet,G}$ and $\mathbf{q}_{\bullet,G}$:
$$\mathbb{F} = \left\{ f_{\bullet,G} : G \in \mathcal{P}(\mathbb{R}^d) \right\}, \ \mathbb{F}_q = \left\{ \mathbf{q}_{\bullet,G} : G \in \mathcal{P}(\mathbb{R}^d) \right\}.$$

3. For $S \subset \mathbb{R}^d$ and $M > 0$, let S^M denote the M-enlargement of a set S:
$$S^M = \{\mathbf{x} \in \mathbb{R}^d : \mathfrak{d}_S(\mathbf{x}) \leq M\}$$
for $\mathfrak{d}_S(\mathbf{x}) = \inf_{s \in S} \|\mathbf{x} - s\|$.

4. Define the semi-norms

$$\|f_{\bullet,G} - f_{\bullet,H}\|_{\infty,S^M} := \max_{1\le k\le K} \sup_{\hat{\boldsymbol{\theta}}\in S^M} \left| f_{k,G}(\hat{\boldsymbol{\theta}}) - f_{k,H}(\hat{\boldsymbol{\theta}}) \right|$$
$$\|\mathbf{q}_{\bullet,G} - \mathbf{q}_{\bullet,H}\|_{\infty,S^M} := \max_{1\le k\le K} \sup_{\hat{\boldsymbol{\theta}}\in S^M} \|\mathbf{q}_{k,G}(\hat{\boldsymbol{\theta}}) - \mathbf{q}_{k,H}(\hat{\boldsymbol{\theta}})\|$$

where $\|\cdot\|$ denotes a vector ℓ_2 norm.

C.3 Proof of Theorem 4.1

The overall concentration strategy follows the η-net approach of Soloff et al. [2025, Theorem 7]; the key new ingredient is the transfer between $f_{k,G}$ and the variance-weighted densities $\tilde{f}_{k,G,i}$ via the condition number τ.

By the conditions on Σ_k, we have

$$\underline{s} f_{k,G} \le \tilde{f}_{k,G,i} \le \bar{s} f_{k,G}, \quad i \le d, k \le K$$

hence

$$\frac{\underline{s} f_{k,G}}{\bar{s} f_{k,\hat{G}}} \le \frac{\tilde{f}_{k,G,i}}{\tilde{f}_{k,\hat{G},i}} \le \frac{\bar{s} f_{k,G}}{\underline{s} f_{k,\hat{G}}}, \quad i \le d, k \le K$$

for any measures $\hat{G}$ and G. This leads to

$$\log\frac{\bar{s}}{\underline{s}} + \ell_K(\hat{G}) - \ell_K(G_0) \ge \frac{1}{K}\sum_{k=1}^{K} \log\frac{\tilde{f}_{k,\hat{G},i}}{\tilde{f}_{k,G_0,i}} \ge \log\frac{\underline{s}}{\bar{s}} + \ell_K(\hat{G}) - \ell_K(G_0), \quad i \le d.$$

Therefore, we only need to prove the bound for $\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0})$ as

$$\left\{ \prod_{k=1}^{K} \frac{\tilde{f}_{k,G,i}(\underline{\hat{\boldsymbol{\theta}}}_k)}{\tilde{f}_{k,G_0,i}(\underline{\hat{\boldsymbol{\theta}}}_k)} \ge \exp\left(-K(t - \log\tau)\right) \right\} \subseteq \left\{ \prod_{k=1}^{K} \frac{f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)} \ge \exp(-Kt) \right\}$$

for any $t \in \mathbb{R}$ and $i = 1, ..., d$.

Given the approximation (5) in the likelihood, we acknowledge that for $t > 0$,

$$\begin{aligned}
&\mathbb{P}\Big(\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq C_{d,\underline{s},\bar{s}} t\varepsilon_M\Big) \\
&\quad = \mathbb{P}\left[\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq C_{d,\underline{s},\bar{s}} t\varepsilon_M, \prod_{k=1}^{K} \frac{f_{k,\hat{G}}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)} \geq \exp\left(-C_{d,\underline{s},\bar{s}} K\varepsilon_M^2\right)\right] \\
&\quad \leq \mathbb{P}\left[\prod_{k=1}^{K} \frac{f_{k,\hat{G}}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)} \geq \exp\left(-C_{d,\underline{s},\bar{s}} K\varepsilon_M^2\right)\right].
\end{aligned}$$

Suppose for some $\tilde{\gamma}_K > 0$, we have

$$\prod_{k=1}^{K} \frac{f_{k,\hat{G}}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)} \geq \exp\left((\beta-\alpha)K\tilde{\gamma}_K^2\right) \text{ for some } 0 < \beta < \alpha < 1$$

and fix $t > 0$. Take $\{f_{\bullet,H_j}\}_{j=1}^N \subset \mathbb{F}$ to be an η-net of $\mathbb{F}$ under $\|\cdot\|_{\infty,S^M}$ for covering number $N := N(\eta, \mathbb{F}, \|\cdot\|_{\infty,S^M})$. For each j, let H_{0j} be a distribution satisfying

$$\|f_{\bullet,H_{0j}} - f_{\bullet,H_j}\|_{\infty,S^M} \leq \eta \quad \& \quad \bar{h}(f_{\bullet,H_{0j}}, f_{\bullet,G_0}) \geq t\tilde{\gamma}_K$$

and $J = \{j \in [N] : H_{0j} \text{ exists}\}$. By construction of the η-net, there is $j^* \in [N]$ such that

$$\|f_{\bullet,H_{j^*}} - f_{\bullet,\hat{G}}\|_{\infty,S^M} \leq \eta.$$

On the event $\{\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq t\gamma_K\}$, the NPMLE $\hat{G}$ acts as a witness that $j^* \in J$, so by the triangle inequality

$$\|f_{\bullet,H_{0j^*}} - f_{\bullet,\hat{G}}\|_{\infty,S^M} \leq 2\eta.$$

This gives

$$f_{k,\hat{G}}(\hat{\underline{\boldsymbol{\theta}}}_k) \leq \begin{cases} f_{k,H_{0j^*}}(\hat{\underline{\boldsymbol{\theta}}}_k) + 2\eta, & \text{if } \hat{\underline{\boldsymbol{\theta}}}_k \in S^M \\ (2\pi\underline{s})^{-d/2}, & \text{otherwise.} \end{cases}$$

Defining $v(\mathbf{x}) := \eta 1_{\mathbf{x}\in S^M} + \eta \left(\frac{M}{\mathfrak{d}_S(\mathbf{x})}\right)^{d+1} 1_{\mathbf{x}\notin S^M}$, we have

$$\begin{aligned}
&\exp\left\{(\beta-\alpha)Kt^2\tilde{\gamma}_K^2\right\} \\
&\quad \leq \max_{j\in J}\left[\prod_{k=1}^{K} \frac{f_{k,H_{0j}}(\hat{\underline{\boldsymbol{\theta}}}_k) + 2v(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\hat{\underline{\boldsymbol{\theta}}}_k)}\right] \cdot \left[\prod_{k:\hat{\underline{\boldsymbol{\theta}}}_k \notin S^M} \frac{(2\pi\underline{s})^{-d/2}}{2v(\hat{\underline{\boldsymbol{\theta}}}_k)}\right]
\end{aligned}$$

on the event $\{\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq t\tilde{\gamma}_K\}$. Hence

$$\mathbb{P}\left(\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq t\tilde{\gamma}_K\right) \leq \mathbb{P}\left[\max_{j\in J}\prod_{k=1}^{K}\frac{f_{k,H_{0j}}(\underline{\hat{\boldsymbol{\theta}}}_k)+2v(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)} \geq \exp(-\alpha K t^2\tilde{\gamma}_K^2)\right]$$
$$+\mathbb{P}\left[\prod_{k:\underline{\hat{\boldsymbol{\theta}}}_k\notin S^M}\frac{(2\pi\underline{s})^{-d/2}}{2v(\underline{\hat{\boldsymbol{\theta}}}_k)} \geq \exp(\beta K t^2\tilde{\gamma}_K^2)\right].$$

Observe that by Markov's inequality, the second term is bounded by

$$\exp\left(-\frac{\beta K t^2\tilde{\gamma}_K^2}{2\log K}\right)\mathbb{E}\left\{\prod_{k:\underline{\hat{\boldsymbol{\theta}}}_k\notin S^M}\frac{(2\pi\underline{s})^{-d/2}}{2v(\underline{\hat{\boldsymbol{\theta}}}_k)}\right\}^{\frac{1}{2\log K}}$$
$$= \exp\left(-\frac{\beta K t^2\tilde{\gamma}_K^2}{2\log K}\right)\mathbb{E}\left[\prod_{k=1}^{K}\left\{\frac{(2\pi\underline{s})^{-\frac{d/2}{d+1}}\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)}{(2\eta)^{1/(d+1)}M}\right\}^{1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M)}}\right]^{\frac{d+1}{2\log K}}.$$

Using similar arguments in Soloff et al. [2025, Proof of Theorem 7], we can obtain the following upper bound for the Hellinger accuracy:

$$\mathbb{P}\left[\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq t\tilde{\gamma}_K\right] \leq N(\eta, \mathbb{F}, \|\cdot\|_{\infty,S^M})\exp\left\{(\alpha-1)\frac{Kt^2\tilde{\gamma}_K^2}{2}+KC_d\sqrt{\eta\,\mathrm{Vol}(S^M)}\right\}$$
$$+\exp\left(-\frac{\beta K t^2\tilde{\gamma}_K^2}{2\log K}+\kappa^\lambda\underline{s}^{-\frac{d\lambda/2}{d+1}}\sum_{k=1}^{K}\mathbb{E}\left[\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)^\lambda 1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M}\right]\right)$$

for $\kappa := \left(\underline{s}^{\frac{d/2}{d+1}}\eta^{\frac{1}{d+1}}M\right)^{-1}$ and $\lambda := \frac{d+1}{2\log K}$. Lemma D.7 gives us

$$\mathbb{E}\left[\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)^\lambda 1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M}\right] \leq C_d M^{d+\lambda-2}\bar{s}^{1-d/2}e^{-M^2/(8\bar{s})}+M^\lambda\left(\frac{2\mu_{k,q}(S,G_0)}{M}\right)^q$$

whenever $\frac{d+1}{2(1\wedge q)} \leq \log K$. Taking $M \geq \sqrt{8\bar{s}\log K}$, we have $e^{-M^2/(8\bar{s})} \leq 1/K$ so

$$\kappa^\lambda\sum_{k=1}^{K}\mathbb{E}\left[\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)^\lambda 1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M}\right] \leq (\kappa M)^\lambda\left\{C_d M^{d-2}\bar{s}^{1-d/2}+\sum_{k=1}^{K}\left(\frac{2\mu_{k,q}(S,G_0)}{M}\right)^q\right\}.$$

Noting $(\kappa M)^\lambda = \left(\underline{s}^{d/2}\eta\right)^{-1/(2\log K)}$, choose $\eta = \frac{1}{K^2\underline{s}^{d/2}}$ so

$$\left(\underline{s}^{d/2}\eta\right)^{-1/(2\log K)} = \left(\frac{4}{K^2}\right)^{-\frac{1}{2\log K}} \leq e$$

Also, Lemma E.3 applies as

$$\log N(\eta, \mathbb{F}, \|\cdot\|_{\infty, S^M}) \leq_d N(a, S^{M+a}) \log^{d+1} c_{d,\underline{s},\bar{s},\bar{s}'} K$$

for $a = \sqrt{\frac{2\bar{s}}{\underline{n}} \log \frac{1}{\eta}} = \sqrt{\frac{4\bar{s}}{\underline{n}} \log \underline{s}^{d/2} K}$. Since $a \in \left(\sqrt{\frac{2\bar{s}}{\underline{n}} \log \underline{s}^{d/2} K}, \sqrt{\frac{6\bar{s}}{\underline{n}} \log \underline{s}^{d/2} K}\right) =: (\underline{a}, \bar{a})$, we have

$$\begin{aligned} N(a, (S^M)^a) \leq N(\underline{a}, S^{M+\bar{a}}) &\leq_d \mathrm{Vol}((S^{M+\bar{a}})^{\underline{a}/2})/\underline{a}^d \\ &\leq_{d,\underline{s},\bar{s}} \mathrm{Vol}(S^{2M}) \left(\frac{\underline{n}}{\log K}\right)^{d/2} \end{aligned} \tag{19}$$

as $\bar{a} \ll M$ in large enough $\underline{n}$. So, aggregating the rates,

$$\log N \leq_d \mathrm{Vol}(S^{2M}) \underline{n}^{d/2} \log^{d/2+1} c_{d,\underline{s}} K.$$

Combining our findings, we have

$$\begin{aligned} &\mathbb{P}\left(\bar{h}(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq t\tilde{\gamma}_K\right) \\ &\leq \exp\left\{-(1-\alpha)\frac{Kt^2\tilde{\gamma}_K^2}{2} + C_d \mathrm{Vol}(S^{2M}) \underline{n}^{d/2} \log^{d/2+1} c_{d,\underline{s}} K + C_{d,\underline{s}} \sqrt{\mathrm{Vol}(S^M)}\right\} \\ &\quad + \exp\left[-\frac{\beta K t^2 \tilde{\gamma}_K^2}{2\log K} + e\left\{C'_{d,\bar{s}} M^{d-2} + \inf_{q \geq \frac{d+1}{2\log K}} \sum_{k=1}^K \left(\frac{2\mu_{k,q}(S, G_0)}{M}\right)^q\right\}\right] \end{aligned}$$

for any $t > 1$. Now, the choice of $\varepsilon_M^2 = \varepsilon_M^2(S, G_0)$ in (4) satisfies

$$\begin{aligned} &\max\left\{\mathrm{Vol}(S^{2M}) \underline{n}^{d/2} \log^{d/2+1} c_{d,\underline{s}} K, \sqrt{\mathrm{Vol}(S^M)}, M^{d-2}, K \log K \inf_{q \geq \frac{d+1}{2\log K}} \sum_{k=1}^K \left(\frac{2\mu_{k,q}(S, G_0)}{M}\right)^q\right\} \\ &\leq_{d,\underline{s}} K\varepsilon_M^2(S, G_0). \end{aligned}$$

If we then take $\tilde{\gamma}_K = \sqrt{\frac{C_{d,s}}{(1-\alpha)\wedge\beta}} \varepsilon_M(M, S, G_0)$, then by the choice of ε_M above, we have $\varepsilon_M^2 \geq \mathrm{Vol}(S^1) \frac{M^d \underline{n}^{d/2} \log^{d/2+1} K}{K} \geq_d \frac{\underline{n}^{d/2} \log^{d+1} K}{K}$ hence

$$\mathbb{P}\left(\bar{h}^2(f_{\bullet,\hat{G}}, f_{\bullet,G_0}) \geq_{d,\underline{s},\bar{s}} t^2\tilde{\gamma}_K^2\right) \leq 2\exp\left(-\frac{t^2 K \varepsilon_M^2}{\log K}\right) \leq 2\exp\left(-t^2 \underline{n}^{d/2} \log^d K\right).$$

Finally, integrate to obtain the upper bound for $\mathbb{E}\left[\frac{\bar{h}^2(f_{\bullet,\hat{G}}, f_{\bullet,G_0})}{\tilde{\gamma}_K^2}\right]$. This completes the proof.

Proof of Corollary 4.1.2. The first claim follows from the compact-support specialization in Saha and Guntuboyina [2020, Corollary 2.2], noting that the client-scaled sets S_k ensure the entropy term remains $\mathcal{O}(\log^{d+1} K/K)$. Next, using the Lipschitz property of $\mathfrak{d}_S$ with the definition (8):

$$\mu_{k,q}(S_k, G_0) \leq \mathfrak{d}_S(\mathbf{0}) + \mu_q(\{\mathbf{0}\}, G_0) \ \ \forall \text{ compact } S$$

where $\mu_q(\{\mathbf{0}\}, G_0) = \mathbb{E}_{\boldsymbol{\theta}\sim G_0}[\|\boldsymbol{\theta}\|^q]^{1/q}$. For large enough K such that $c > \frac{d+1}{2\log K}$, we have:

$$\inf_{q \geq \frac{d+1}{2\log K}} \frac{1}{K}\sum_{k=1}^{K}\left(\frac{2\mu_{k,q}(S_k, G_0)}{M}\right)^q \leq_c \left(\frac{1}{M}\right)^c$$

for all $k \leq K$. Choosing

$$M = (\log K)^{-\frac{d/2}{c+d}}\left\{\frac{KR^c}{\mathrm{Vol}(S^1)}\right\}^{\frac{1}{c+d}},$$

we have the conclusion as the first term comes from Saha and Guntuboyina [2020, (2.10)]. $\square$

C.4 Oracle denoising inequality: proof of Theorem 4.2

Throughout the proof we will group sequences of the form $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K$ into $K \times d$ matrices Θ, so that, for instance, the regret $\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\|\hat{\boldsymbol{\theta}}_k^v - \boldsymbol{\theta}_k^o\|^2$ in the statement of the theorem may be rewritten as the expected squared Frobenius norm $\frac{1}{K}\mathbb{E}\|\hat{\Theta}^v - \Theta^v\|_F^2$.

C.4.1 Regularizing the Bayes rule

Before evaluating $\hat{\Theta}^v$, we verify that the likelihood is not decaying at each of the observations. By (12), for any fixed $j \in [K]$,

$$\begin{aligned}\prod_{k=1}^{K} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k) &\geq \prod_{k=1}^{K} f_{k,K^{-1}\delta_{\underline{\hat{\boldsymbol{\theta}}}_k}+(1-K^{-1})\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k) \\ &\geq K^{-1}|2\pi\Sigma_j|^{-1/2}(1-K^{-1})^{K-1}\prod_{k:k\neq j} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k).\end{aligned}$$

Cancelling terms for $k \in [K] \setminus \{j\}$, we conclude

$$f_{j,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_j) \geq \frac{\inf_{\mathbf{u}}|2\pi\Sigma_j(\mathbf{u})|^{-1/2}}{eK} = \frac{(2\pi\bar{s})^{-d/2}}{eK} =: \rho_0. \tag{20}$$

Given this, Saha and Guntuboyina [2020], Soloff et al. [2025] adopt the regularized empirical Bayes and oracle Bayes rules

$$\hat{\boldsymbol{\theta}}^v_{k,\rho} = \frac{\hat{\boldsymbol{\theta}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,\hat{G}}(\hat{\boldsymbol{\theta}}_k)}{\sqrt{n_k} f_{k,\hat{G}}(\hat{\boldsymbol{\theta}}_k) \vee \rho} \tag{21}$$

$$\boldsymbol{\theta}^o_{k,\rho} = \frac{\hat{\boldsymbol{\theta}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,G_0}(\hat{\boldsymbol{\theta}}_k)}{\sqrt{n_k} f_{k,G_0}(\hat{\boldsymbol{\theta}}_k) \vee \rho}.$$

By the lower bound (20) we know that $\hat{\Theta}^v_\rho = \hat{\Theta}^v$ when $\rho \le \rho_0$ hence

$$\|\hat{\Theta}^v - \Theta^v\|_F = \|\hat{\Theta}^v_\rho - \Theta^v\|_F \le \|\hat{\Theta}^v_\rho - \Theta^v_\rho\|_F + \|\Theta^v_\rho - \Theta^v\|_F. \tag{22}$$

The first term $\|\hat{\Theta}^v_\rho - \Theta^v_\rho\|_F$ represents the regret between regularized rules, which prevents the denominator in Tweedie's formula from blowing up. The second term represents the cost of introducing a small amount of regularization in the oracle Bayes rule.

C.4.2 Regularization error of oracle Bayes

Let us first consider the second term $\|\Theta^v_\rho - \Theta^v\|_F$ on the RHS of the bound (22). The argument of Soloff et al. [2025, Proof of Theorem 9] does not apply here because our marginal densities are scaled mixtures over kernels that depend on both $\boldsymbol{\theta}$ and $\Sigma_k(\boldsymbol{\theta})$. We therefore develop a modified analysis that routes through the variance-weighted functional $\tilde{\Delta}_k$ (Lemmas D.3 and D.4), extending Saha and Guntuboyina [2020, Lemmas 4.3 and F.6] to our setting. Define

$$\tilde{\Delta}_k(G, \rho) := \int \left(1 - \frac{f_{k,G}}{f_{k,G} \vee \rho}\right)^2 \frac{\|\mathbf{q}_{k,G_0}\|^2}{f_{k,G}}.$$

Then, for any compact $S \subseteq \mathbb{R}^d$, letting $S_{\underline{n}} := \frac{1}{\sqrt{\underline{n}}} S$:

$$\begin{aligned}\mathbb{E}\|\Theta^v_\rho - \Theta^v\|_F^2 &= \sum_{k=1}^K \frac{1}{n_k} \mathbb{E}\left\| \frac{\mathbf{q}_{k,G_0}(\hat{\boldsymbol{\theta}}_k)}{f_{k,G_0}(\hat{\boldsymbol{\theta}}_k) \vee \rho} - \frac{\mathbf{q}_{k,G_0}(\hat{\boldsymbol{\theta}}_k)}{f_{k,G_0}(\hat{\boldsymbol{\theta}}_k)} \right\|^2 \le \sum_{k=1}^K \frac{1}{n_k} \tilde{\Delta}_k (G_0, \rho) \\ &\lesssim_{d,\underline{s},\bar{s}} \frac{K}{\underline{n}} \left\{ N\left(\frac{4}{\sqrt{\underline{n}} L(\rho)}, S_{\underline{n}}\right) \rho L(\rho)^d + G_0(S^c_{\underline{n}}) \right\}\end{aligned}$$

by Lemmas D.3 and D.4 for $L(\rho) := \sqrt{\underline{s} \log \frac{1}{(2\pi\bar{s})^d \rho^2}}$ whenever $\rho \le (2\pi\bar{s})^{-d/2} e^{-8/\underline{s}}$. Choosing

$$\rho = \frac{(2\pi\bar{s})^{-d/2}}{e^{1+8/\underline{s}} K} \tag{23}$$

and $S = S^M$, we have $L(\rho) = \sqrt{2\underline{s}\log(Ke^{1+8/\underline{s}})}$ hence

$$\begin{aligned}
\frac{\underline{n}}{K}\mathbb{E}\|\Theta_\rho^v - \Theta^v\|_F^2 &\leq_{d,\underline{s},\bar{s}} N\left(\frac{1}{\sqrt{2\underline{s}\,\underline{n}\log K}}, S_{\underline{n}}^M\right)\frac{\log^{d/2} K}{K} + G_0((S_{\underline{n}}^M)^c) \\
&\leq_d \mathrm{Vol}(S_{\underline{n}}^{2M})\,(2\underline{s}\,\underline{n}\log K)^{d/2}\,\frac{\log^{d/2} K}{K} + G_0((S_{\underline{n}}^M)^c) \\
&\leq_{d,\underline{s}} \mathrm{Vol}\left(S^1\right) M^d \frac{\log^d K}{K} + \inf_{q \geq \frac{d+1}{2\log K}} \left(\frac{2\mu_q}{M}\right)^q \\
&\qquad \text{for } \mu_q := \mathbb{E}_{G_0}[\mathfrak{d}_{S_{\underline{n}}}(\boldsymbol{\theta})^q]^{1/q}
\end{aligned}$$

by applying Saha and Guntuboyina [2020, Lemma F.6]. The second term is relevant with the second term in (8). Therefore,

$$\frac{1}{K}\mathbb{E}\|\Theta_\rho^v - \Theta^v\|_F^2 \leq_{d,\underline{s},\bar{s}} \mathrm{Vol}\left(S^1\right) M^d \frac{\log^d K}{\underline{n}K} + \frac{1}{\underline{n}} \inf_{q \geq \frac{d+1}{2\log K}} \left(\frac{2\mu_q}{M}\right)^q \lesssim \frac{\varepsilon_M^2(S_\bullet, G_0)}{\underline{n}}.$$

C.4.3 Regret of regularized rules

Now we consider

$$\left\|\hat{\Theta}_\rho^v - \Theta_\rho^v\right\|_F^2 = \sum_{k\leq K}\frac{1}{n_k}\left\|\frac{\mathbf{q}_{k,\hat{G}}(\hat{\boldsymbol{\theta}}_k)}{f_{k,\hat{G}}(\hat{\boldsymbol{\theta}}_k)\vee\rho} - \frac{\mathbf{q}_{k,G}(\hat{\boldsymbol{\theta}}_k)}{f_{k,G}(\hat{\boldsymbol{\theta}}_k)\vee\rho}\right\|^2$$

on the RHS of the bound (22). First, we will introduce some additional notation. For $\delta > 0$ define $A_\delta = \left\{\max_{i\leq d}\bar{h}^2(\tilde{f}_{\bullet,\hat{G},i}, \tilde{f}_{\bullet,G_0,i}) \vee \bar{h}^2\left(f_{\bullet,\hat{G}}, f_{\bullet,G_0}\right) \leq \delta\right\}$. Given a compact set $S \subset \mathbb{R}^d$, define another metric

$$m^S(G,H) := \frac{1}{\sqrt{\underline{n}}}\max_{k\in[K]}\sup_{\mathbf{z}\in\mathbb{R}^d:\mathfrak{d}_{S_k}(\mathbf{z})\leq M}\left\|\frac{\mathbf{q}_{k,G}(\mathbf{z})}{f_{k,G}(\mathbf{z})\vee\rho} - \frac{\mathbf{q}_{k,H}(\mathbf{z})}{f_{k,H}(\mathbf{z})\vee\rho}\right\|$$

for $S_k := \frac{1}{\sqrt{n_k}}S$. Let $G_1,\ldots,G^{(N)}$ denote a minimal η^*-covering of

$$\mathcal{H}_\delta := \left\{G : \max_{i\leq d}\bar{h}^2(\tilde{f}_{\bullet,G,i}, \tilde{f}_{\bullet,G_0,i}) \vee \bar{h}^2(f_{\bullet,G}, f_{\bullet,G_0}) \leq \delta\right\} \tag{24}$$

in the metric m^S for $N := N(\eta^*, \mathcal{H}_\delta, m^S)$. For $j \in [N]$ similarly define a $K \times d$ matrix $\hat{\Theta}_\rho^{v,(j)}$ where the i^{th} row is given by $\frac{\hat{\boldsymbol{\theta}}_k}{\sqrt{n_k}} + \frac{\mathbf{q}_{k,G^{(j)}}(\hat{\boldsymbol{\theta}}_k)}{\sqrt{n_k}f_{k,G^{(j)}}(\hat{\boldsymbol{\theta}}_k)\vee\rho}$. We bound the regret as $\|\hat{\Theta}_\rho^v - \Theta_\rho^v\|_F \leq \sum_{t=1}^4 Q_t$, where

$$
\begin{aligned}
Q_1 &:= \|\hat{\Theta}^v_\rho - \Theta^v_\rho\|_F 1_{A^c_\delta} \\
Q_2 &:= \left(\|\hat{\Theta}^v_\rho - \Theta^v_\rho\|_F - \max_{j\in[N]}\|\hat{\Theta}^{v,(j)}_\rho - \Theta^v_\rho\|_F\right)_+ 1_{A_\delta} \\
Q_3 &:= \max_{j\in[N]}\left(\|\hat{\Theta}^{v,(j)}_\rho - \Theta^v_\rho\|_F - \mathbb{E}\|\hat{\Theta}^{v,(j)}_\rho - \Theta^v_\rho\|_F\right)_+ \\
Q_4 &:= \max_{j\in[N]} \mathbb{E}\|\hat{\Theta}^{v,(j)}_\rho - \Theta^v_\rho\|_F.
\end{aligned}
\tag{25}
$$

Bounding $\mathbb{E}(Q_1^2)$. By the shrinkage Tweedie's formula in (21),

$$
\begin{aligned}
\|\hat{\boldsymbol{\theta}}^v_{k,\rho} - \boldsymbol{\theta}^o_{k,\rho}\|^2 &= \frac{1}{n_k}\left\|\frac{\mathbf{q}_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho} - \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho}\right\|^2 \\
&\le \frac{1}{n_k}\left\|\frac{\mathbf{q}_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho}\right\|^2 + \frac{1}{n_k}\left\|\frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho}\right\|^2.
\end{aligned}
$$

Lemma D.1 provides

$$
\mathbb{E}(Q_1^2) \lesssim_{\bar{s}} \frac{K}{\underline{n}}\log\frac{1}{(2\pi\underline{s})^d\rho^2} \times \Pr(A^c_\delta) \le \frac{K\log(K^2e^{1+8/\underline{s}})}{\underline{n}}\Pr(A^c_\delta).
$$

By Corollary 4.1.2, there is a constant $C_d > 0$ such that $\delta = C_d\varepsilon^2_M(S_\bullet, G_0)$ satisfies $\Pr(A^c_\delta) \le 2\exp\left(-\log^d K\right)$. Hence

$$
\frac{1}{K}\mathbb{E}(Q_1^2) \lesssim_{\bar{s}} \frac{2\log(Ke^{1/2+4/\underline{s}})}{\underline{n}\exp(\log^d K)}.
$$

Bounding $\mathbb{E}\left(Q_2^2\right)$. Observe that

$$
\begin{aligned}
Q_2^2 &\le 1_{A_\delta}\min_{j\in[N]}\|\hat{\Theta}^v_\rho - \hat{\Theta}^{v,(j)}_\rho\|^2_F \\
&= 1_{A_\delta}\min_{j\in[N]}\sum_{k=1}^K\frac{1}{n_k}\left\|\frac{\mathbf{q}_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho} - \frac{\mathbf{q}_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k)\vee\rho}\right\|^2.
\end{aligned}
$$

On A_δ, we may take j such that $m^S(\hat{G}, G^{(j)}) \le \eta^*$. For each k, consider two cases, where $\underline{\hat{\boldsymbol{\theta}}}_k \in S^M_k$ and where $\underline{\hat{\boldsymbol{\theta}}}_k \notin S^M_k$. When $\underline{\hat{\boldsymbol{\theta}}}_k \in S^M_k$ bound the above by the supremum over all $\underline{\hat{\boldsymbol{\theta}}}_k \in S^M_k$. When $\underline{\hat{\boldsymbol{\theta}}}_k \notin S^M_k$ bound the regularized rules as before. This yields

$$
Q_2^2 \lesssim_d 1_{A_\delta}\left[K(\eta^*)^2 + \#\{k : \underline{\hat{\boldsymbol{\theta}}}_k \notin S^M_k\}\frac{\log(K^2e^{1+8/\underline{s}})}{\underline{n}}\right]
$$

hence

$$\frac{\mathbb{E}(Q_2^2)}{K} \leq_d (\eta^*)^2 + \frac{\log(K^2 e^{1+8/\underline{s}})}{\underline{n}K} \sum_{k=1}^K \Pr\left(\mathfrak{d}_{S_k}(\underline{\hat{\boldsymbol{\theta}}}_k) \geq M\right).$$

By Lemma D.7, taking $\lambda \downarrow 0$,

$$\begin{aligned}
\frac{\mathbb{E}(Q_2^2)}{K} &\leq_{d,\underline{s}} (\eta^*)^2 + \frac{\log(K^2 e^{1+8/\underline{s}})}{\underline{n}K} \left\{ M^{d-2} e^{-M^2/(8\bar{s})} + \sum_{k=1}^K \inf_{q \geq \frac{d+1}{2\log K}} \left(\frac{2\mu_{k,q}(S_k, G_0)}{M}\right)^q \right\} \\
&\leq (\eta^*)^2 + \frac{\log(K^2 e^{1+8/\underline{s}})}{\underline{n}K} \left\{ \frac{M^{d-2}}{K} + \sum_{k=1}^K \inf_{q \geq \frac{d+1}{2\log K}} \left(\frac{2\mu_{k,q}(S_k, G_0)}{M}\right)^q \right\}
\end{aligned}$$

given that $M \geq \sqrt{8\bar{s}\log K}$.

Bounding $\mathbb{E}Q_3^2$. Fix $j \in [N]$. Then,

$$\begin{aligned}
\|\hat{\boldsymbol{\theta}}_{k,\rho}^{v,(j)} - \boldsymbol{\theta}_{k,\rho}^o\|^2 &\leq \frac{1}{n_k} \left\| \frac{\mathbf{q}_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} - \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \right\|^2 \\
&\leq \frac{2}{n_k} \left(\left\| \frac{\mathbf{q}_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \right\|^2 + \left\| \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \right\|^2 \right).
\end{aligned}$$

By Lemma D.1,

$$\|\hat{\boldsymbol{\theta}}_{k,\rho}^{v,(j)} - \boldsymbol{\theta}_{k,\rho}^o\|^2 \leq -\frac{4\bar{s}}{n_k} \log(2\pi\underline{s})^d \rho^2 \leq_{d,\underline{s},\bar{s}} \frac{\log(K e^{1+8/\underline{s}})}{\underline{n}} =: r^2.$$

Let $V_{k,j} = \frac{\left\|\hat{\boldsymbol{\theta}}_{k,\rho}^{v,(j)} - \boldsymbol{\theta}_{k,\rho}^o\right\|}{r}$, so $\|\hat{\Theta}_\rho^{v,(j)} - \Theta_\rho^v\|_F^2 = r^2 \|\boldsymbol{V}_j\|^2$ for $\boldsymbol{V}_j := (V_{1,j}, \ldots, V_{K,j})$. Since $V_{k,j}$ are independent random variables in $[0,1]$, it has an exponentially decreasing tail probability with

$$\Pr\left(\|\boldsymbol{V}_j\| \geq \mathbb{E}\|\boldsymbol{V}_j\| + t\right) \leq e^{-t^2/2}$$

for some $c > 0$ independent of $t \in \mathbb{R}$. Thus

$$\Pr\left(Q_3 \geq s\right) \leq N \exp\left(-C_{d,\underline{s},\bar{s}} \frac{s^2}{r^2}\right), \quad s > 0.$$

Therefore,

$$\frac{1}{K}\mathbb{E}(Q_3^2) \leq_d \frac{r^2}{K} \log(eN) = \log(eN) \frac{\log(K e^{1+8/\underline{s}})}{K\underline{n}}. \tag{26}$$

Bounding $\mathbb{E}Q_4^2$. Again

$$\begin{aligned}
\mathbb{E}\|\hat{\Theta}_\rho^{v,(j)} - \Theta_\rho^v\|_F &\le \sqrt{\mathbb{E}\|\hat{\Theta}_\rho^{v,(j)} - \Theta_\rho^v\|_F^2} \\
&\le \sqrt{\frac{1}{\underline{n}}\sum_{k=1}^K \mathbb{E}_{\underline{\hat{\boldsymbol{\theta}}}_k \sim f_{k,G_0}} \left\| \frac{\mathbf{q}_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G^{(j)}}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} - \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \right\|^2}.
\end{aligned}$$

Then, Theorem D.6 gives us

$$\frac{1}{K}\left(\mathbb{E}\|\hat{\Theta}_\rho^{v,(j)} - \Theta_\rho^v\|_F\right)^2 \lesssim_{d,\underline{s},\bar{s}} \frac{1}{\underline{n}}\left[d \max\left\{\log^3(Ke^{1+8/\underline{s}}), \log\frac{1}{\delta}\right\}\right]\delta.$$

Recall our choice

$$\delta = \varepsilon_M^2(S_\bullet, G_0) \gtrsim_{d,\underline{s},\bar{s}} \frac{\mathrm{Vol}(S^1)M^d \log^{d/2+1} K}{K} \ge \frac{1}{K}$$

hence we obtain

$$\frac{\mathbb{E}(Q_4^2)}{K} \lesssim_{d,\underline{s},\bar{s}} \frac{\log^3 K}{\underline{n}}\varepsilon_M^2(S_\bullet, G_0).$$

Metric Entropy for m^S For any $G, \tilde{G} \in \mathcal{P}(\mathbb{R}^d)$, note that

$$\begin{aligned}
\sqrt{\underline{n}}m^S(G,\tilde{G}) &= \max_{k\in[K]} \sup_{\mathbf{x}:\mathfrak{d}_{S_k}(\mathbf{x})\le M} \left\| \frac{\mathbf{q}_{k,G}(\mathbf{x})}{f_{k,G}(\mathbf{x}) \vee \rho} - \frac{\mathbf{q}_{k,\tilde{G}}(\mathbf{x})}{f_{k,\tilde{G}}(\mathbf{x}) \vee \rho} \right\| \\
&\le \max_{k\in[K]} \sup_{\mathbf{x}:\mathfrak{d}_{S_k}(\mathbf{x})\le M} \frac{\|\mathbf{q}_{k,G}(\mathbf{x})\|}{f_{k,G}(\mathbf{x}) \vee \rho} \cdot \frac{\left| f_{k,\tilde{G}}(\mathbf{x}) \vee \rho - f_{k,G}(\mathbf{x}) \vee \rho \right|}{f_{k,\tilde{G}}(\mathbf{x}) \vee \rho} \\
&\quad + \max_{k\in[K]} \sup_{\mathbf{x}:\mathfrak{d}_{S_k}(\mathbf{x})\le M} \frac{\left\| \mathbf{q}_{k,G}(\mathbf{x}) - \mathbf{q}_{k,\tilde{G}}(\mathbf{x}) \right\|}{f_{k,\tilde{G}}(\mathbf{x}) \vee \rho} \\
&\le \sqrt{-\bar{s}\log(2\pi\underline{s})^d\rho^2} \max_{k\in[K]} \sup_{\mathbf{x}:\mathfrak{d}_{S_k}(\mathbf{x})\le M} \frac{\left| f_{k,\tilde{G}}(\mathbf{x}) - f_{k,G}(\mathbf{x}) \right|}{\rho} \\
&\quad + \max_{k\in[K]} \sup_{\mathbf{x}:\mathfrak{d}_{S_k}(\mathbf{x})\le M} \frac{\left\| \mathbf{q}_{k,G}(\mathbf{x}) - \mathbf{q}_{k,\tilde{G}}(\mathbf{x}) \right\|}{\rho}.
\end{aligned} \tag{27}$$

Therefore,

$$m^S(G,\tilde{G}) \le A \max_{k\in[K]} \|f_{k,G} - f_{k,\tilde{G}}\|_{\infty,S_k^M} + B \max_{k\in[K]} \|\mathbf{q}_{k,G} - \mathbf{q}_{k,\tilde{G}}\|_{\infty,S_k^M}$$

for $A := \frac{1}{\rho}\sqrt{-\frac{\bar{s}}{\underline{n}}\log(2\pi\underline{s})^d\rho^2}$ and $B := \frac{1}{\rho\sqrt{\underline{n}}}$. Letting $\eta^* := \eta(A+B)$, we have

$$\log N(\eta^*, \mathcal{P}(\mathbb{R}^d), m^S) \leq \max_{k\in[K]} \log N(\eta/2, \mathbb{F}, \|\cdot\|_{\infty,S_k^M}) + \max_{k\in[K]} \log N(\eta/2, \mathbb{F}_q, \|\cdot\|_{\infty,S_k^M})$$
$$\leq_d \max_{k\in[K]} \log N(a, S_k^{M+a}) \log^{d+1} \frac{c_{d,\underline{s},\bar{s},\bar{s}'}}{\eta}$$

for $a = \sqrt{\frac{2\bar{s}}{\underline{n}}\log\frac{1}{\eta}}$ and small $\eta > 0$ by Lemma E.3. Let $\eta = \rho$ so that

$$\log N(\eta^*, \mathcal{P}(\mathbb{R}^d), m^S) \leq_{d,\underline{s},\bar{s}} \mathrm{Vol}(S^1) M^d \log^{d/2+1} K \tag{28}$$

for

$$\eta^* \leq_{d,\underline{s},\bar{s}} \frac{M}{\sqrt{\underline{n}}}.$$

Final oracle denoising bound To obtain the oracle denoising bound, we aggregate the rates from the error decomposition. The overall risk decomposes as

$$\frac{1}{K}\mathbb{E}\|\hat{\Theta}^v - \Theta^v\|_F^2 \leq \frac{1}{K}\mathbb{E}\|\Theta_\rho^v - \Theta^v\|_F^2 + \frac{1}{K}\sum_{t=1}^4 \mathbb{E}(Q_t^2).$$

Combining the individual bounds:

- $\frac{1}{K}\mathbb{E}\|\Theta_\rho^v - \Theta^v\|_F^2 \lesssim \frac{\varepsilon_M^2(S_\bullet, G_0)}{\underline{n}} \log^{d/2-1} K$ from the worst-case density floor.
- $\frac{1}{K}\mathbb{E}(Q_1^2) \leq_{\bar{s}} \log(Ke^{1+8/\underline{s}})\underline{n}^{-1}\exp(-\log^d K)$ is exponentially suppressed.
- $\frac{1}{K}\mathbb{E}(Q_2^2) \leq_d \frac{M^2}{\underline{n}} + \frac{\log(K^2e^{1+8/\underline{s}})}{K\underline{n}} \left\{\frac{M^{d-2}}{K} + \sum_{k=1}^K \inf_{q\geq \frac{d+1}{2\log K}} \left(\frac{2\mu_{k,q}(S_k,G_0)}{M}\right)^q\right\} \lesssim \varepsilon_M^2(S_\bullet, G_0)$ provided that $\underline{n} \gtrsim KM^{2-d}\log^{-d/2-1} K$ for (8).
- $\frac{1}{K}\mathbb{E}(Q_3^2) \leq_d \frac{\mathrm{Vol}(S^1)M^d\log^{d/2+2} K}{K\underline{n}}$ from the metric entropy of the Hellinger ball.
- $\frac{1}{K}\mathbb{E}(Q_4^2) \leq_{d,\underline{s},\bar{s}} \frac{\log^3(Ke^{1+8/\underline{s}})}{\underline{n}}\varepsilon_M^2(S_\bullet, G_0)$ captures the cost of restricting to the Hellinger-ball neighborhood.

Summing these contributions:

$$\frac{1}{K}\mathbb{E}\|\hat{\Theta}^v - \Theta^v\|_F^2 \leq_{d,\underline{s},\bar{s}} \frac{\log^{3\vee(d/2-1)} K}{\underline{n}} \varepsilon_M^2(S_\bullet, G_0).$$

When G_0 is a Gaussian Mixture. For the case when G_0 is a Gaussian Mixture, we only need to bound the second term in (8). Let $R_k := R/\sqrt{n_k}$ and $S := \mathbb{B}_R(\mathbf{0}_d)$. By the linearity of expectation and the definition of G_0, we have:

$$\mathbb{E}_{G_0}[\mathfrak{d}_{S_k}(\boldsymbol{\theta})^q] = \sum_{j=1}^{k^*} w_j \mathbb{E}_{\boldsymbol{\theta}\sim N(\boldsymbol{a}_j^*,\Gamma_j)}(\|\boldsymbol{\theta}\| - R_k)_+^q.$$

For each component j, let $\boldsymbol{\theta} = \boldsymbol{a}_j^* + \boldsymbol{\epsilon}_j$, where $\boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}_d, \Gamma_j^*)$. Define $\bar{\gamma}^2 := \max_{j\le k^*}\|\Gamma_j^*\|$ and $A^* := \max_{j\le k^*}\|\mathbf{a}_j^*\|$. By the triangle inequality:

$$\|\boldsymbol{\theta}\| \le \|\boldsymbol{a}_j^*\| + \|\boldsymbol{\epsilon}_j\| \le A^* + \|\boldsymbol{\epsilon}_j\|.$$

Thus, the distance to S_k is bounded by:

$$\mathfrak{d}_{S_k}(\boldsymbol{\theta}) \le (A^* + \|\boldsymbol{\epsilon}_j\| - R_k)_+.$$

When $R_k \to \infty$, this term vanishes, and the term of interest also does. We only consider the case $R_k < \infty$, i.e., $R_k = \mathcal{O}(\sqrt{n_k})$. For a conservative upper bound, we observe that for any $R_k \ge 0$:

$$\mathfrak{d}_{S_k}(\boldsymbol{\theta})^q \le (A^* + \|\boldsymbol{\epsilon}_j\|)^q.$$

For $X := \|\boldsymbol{\epsilon}_j\|$, $X - \mathbb{E}X$ is sub-Gaussian with parameter $\bar{\gamma}$ and $\mathbb{E}(X) \le \sqrt{\mathrm{tr}(\Gamma_j)} \le \sqrt{d}\bar{\gamma}$. Using the property of sub-Gaussian moments, for $q \ge 1$:

$$\mathbb{E}(X^q)^{1/q} \le C\sqrt{\bar{\gamma}(d+q)}$$

for some universal constant $C > 0$. Therefore, by Minkowski's inequality:

$$\mathbb{E}[\mathfrak{d}_{S_k}(\boldsymbol{\theta})^q]^{1/q} \le A^* + C\sqrt{\bar{\gamma}(d+q)}.$$

Substituting this back into the mixture sum:

$$\mathbb{E}_{G_0}[\mathfrak{d}_{S_k}(\boldsymbol{\theta})^q] \le \sum_{j=1}^{k^*} w_j \left(A^* + C\sqrt{\bar{\gamma}(d+q)}\right)^q = \left(A^* + C\sqrt{\bar{\gamma}(d+q)}\right)^q$$

Now, consider the ratio with M^q. Let M be chosen such that $M \gtrsim \sqrt{\log K}$ as before. Therefore, the term of interest becomes negligible as K is assumed to diverge.

This completes the proof of Theorem 4.2.

D Technical Results

Proof of Lemma 3.1. The first-order optimality condition for $\hat{G}$ is standard [Soloff et al., 2025, Lemma 1]:

$$\frac{1}{K}\sum_{k=1}^{K}\frac{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)} \leq 1 \ \ \forall G \in \mathcal{P}(\mathbb{R}^d).$$

When $G = \delta_{\boldsymbol{t}}$, we write $D(\hat{G}, G) = D(\hat{G}, \boldsymbol{t})$ and it suffices to check $D(\hat{G}, \boldsymbol{t}) \leq 0$. For the first claim, define $\mathcal{C} := \left\{f_{1,G}(\underline{\hat{\boldsymbol{\theta}}}_1), ..., f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_K); G \in \mathcal{P}(\mathbb{R}^d)\right\} \cup \{0\}$. Observe that

$$\mathcal{C} = \mathrm{conv}(\mathcal{L}) \ \ \text{where} \ \ \mathcal{L} := \left\{\varphi^{(1)}(\underline{\hat{\boldsymbol{\theta}}}_1; \boldsymbol{\theta}), ..., \varphi^{(K)}(\underline{\hat{\boldsymbol{\theta}}}_K; \boldsymbol{\theta}); \boldsymbol{\theta} \in \mathbb{R}^d\right\}.$$

Since $\boldsymbol{\theta} \mapsto \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta})$ is continuous and $\lim_{\|\boldsymbol{\theta}\|\to\infty} \max_{k\leq K} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) = 0$, $\mathcal{L}$ is closed, and by boundedness of the Gaussian likelihood, $\mathcal{L}$ is compact and closed. Hence $\mathcal{C}$ is compact and convex, and $\sum_{k=1}^{K} \log f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)$ is strictly concave and coordinate-wise monotone over $\mathcal{C}$. Thus, NPMLE is attained at a unique (non-zero) boundary point $\partial\mathcal{C}$. By Carathéodory's Theorem, such boundary point can be written as $\sum_{j=1}^{\hat{k}} \hat{w}_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{a}_j)$ for some $\hat{k} \leq K$. The second claim follows directly from Soloff et al. [2025]. □

Proof of Lemma 3.2. With frozen covariances $\mathbf{S}_k$, the conclusion directly follows from Ray and Lindsay [2005, Theorem 1]. □

D.1 Convergence of the pseudo-EM under Lipschitz covariances

Proof of Lemma A.1. Write $\mathbf{P}_k(\mathbf{a}) := n_k \Sigma_k(\mathbf{a})^{-1}$, $\mathbf{A}(\mathbf{a}) := \sum_{k=1}^{K} w_{kj}^{(t)} \mathbf{P}_k(\mathbf{a})$, and $\mathbf{b}(\mathbf{a}) := \sum_{k=1}^{K} w_{kj}^{(t)} \mathbf{P}_k(\mathbf{a})\hat{\boldsymbol{\theta}}_k$, so $\bar{\boldsymbol{\psi}}_j(\mathbf{a}) = \mathbf{A}^{-1}\mathbf{b}$. Since $w_{kj}^{(t)} \geq 0$ and $\sum_{k=1}^{K} w_{kj}^{(t)} = 1$, the output is a weighted average of $\{\hat{\boldsymbol{\theta}}_k\}$, so $\bar{\boldsymbol{\psi}}$ maps $\mathrm{conv}(\hat{\boldsymbol{\theta}}_1, \ldots, \hat{\boldsymbol{\theta}}_K)^{\hat{k}}$ into itself.

The resolvent identity and $\underline{s}\mathbf{I}_d \preceq \Sigma_k \preceq \bar{s}\mathbf{I}_d$ give

$$\|\Sigma_k(\mathbf{a}_1)^{-1} - \Sigma_k(\mathbf{a}_2)^{-1}\| \leq (L_\Sigma/\underline{s}^2)\|\mathbf{a}_1 - \mathbf{a}_2\|. \tag{29}$$

From $\mathbf{A}\bar{\boldsymbol{\psi}}_j = \mathbf{b}$, differencing at two points gives

$$\bar{\boldsymbol{\psi}}_j(\mathbf{a}_1) - \bar{\boldsymbol{\psi}}_j(\mathbf{a}_2) = \mathbf{A}(\mathbf{a}_1)^{-1} \sum_{k=1}^{K} w_{kj}^{(t)} [\mathbf{P}_k(\mathbf{a}_1) - \mathbf{P}_k(\mathbf{a}_2)](\hat{\boldsymbol{\theta}}_k - \bar{\boldsymbol{\psi}}_j(\mathbf{a}_2)).$$

Since $\|\mathbf{A}^{-1}\| \leq \bar{s}/n_j^w$, $\|\mathbf{P}_k(\mathbf{a}_1) - \mathbf{P}_k(\mathbf{a}_2)\| \leq n_k L_\Sigma \|\mathbf{a}_1 - \mathbf{a}_2\|/\underline{s}^2$, and $\|\hat{\boldsymbol{\theta}}_k - \bar{\boldsymbol{\psi}}_j(\mathbf{a}_2)\| \leq R$, we obtain

$\|\bar{\boldsymbol{\psi}}_j(\mathbf{a}_1) - \bar{\boldsymbol{\psi}}_j(\mathbf{a}_2)\| \le L_{\bar{\psi}}\|\mathbf{a}_1 - \mathbf{a}_2\|$. Brouwer fixed-point theorem gives existence; if $L_{\bar{\psi}} < 1$, Banach fixed-point theorem gives uniqueness and linear convergence.

Next, we define the per-atom quantities

$$\chi_j^2 := \sum_{k=1}^{K} w_{kj}^{(t)} \underbrace{\sum_{\ell} \frac{(\underline{\hat{\theta}}_{k,\ell} - \sqrt{n_k} a_{j,\ell}^*)^2}{\sigma_{k,\ell}(\mathbf{a}_j^*)^2}}_{=: \chi_{k,j}^2}, \qquad \kappa_j := \frac{2\sqrt{d}\, \bar{s}^{3/2} L_\Sigma \chi_j}{\underline{s}^2 n_j^w}.$$

At $\mathbf{a}_j^*$ the frozen score vanishes, so $\sum_{k=1}^K w_{kj}^{(t)} \nabla \psi_k(\mathbf{a}_j^*) = \sum_{k=1}^K w_{kj}^{(t)} \mathbf{r}_k(\mathbf{a}_j^*)$ where $r_{k,i}$ collects the $\nabla \sigma_{k,i}$ terms. Since $|\nabla_{\theta_i} \sigma_{k,\ell}|/\sigma_{k,\ell} \le L_\Sigma/(2\underline{s})$ and $|u-1| \le u+1$, each component satisfies $|r_{k,i}| \le \frac{L_\Sigma}{2\underline{s}}(\chi_{k,j}^2 + 1)$, giving

$$\left\| \sum_{k=1}^K w_{kj}^{(t)} \nabla \psi_k(\mathbf{a}_j^*) \right\| \le \frac{\sqrt{d}\, L_\Sigma}{2\underline{s}}(\chi_j^2 + 1). \tag{30}$$

Write $\nabla_{\mathbf{a}} \boldsymbol{f}_j = -\mathbf{A} + \boldsymbol{E}$ where $\boldsymbol{f}_j(\mathbf{a}) := \sum_{k=1}^K w_{kj}^{(t)} \nabla \psi_k(\mathbf{a})$, with $\mathbf{A} \succeq n_j^w/\bar{s}\, \mathbf{I}_d$. By Cauchy-Schwarz, $\sum_{k=1}^K w_{kj}^{(t)} |\underline{\hat{\theta}}_{k,i} - \sqrt{n_k} a_{j,i}^*| \le \sqrt{\bar{s}\chi_j^2}$, so $\|\boldsymbol{E}\|_F \le 2\sqrt{d}\, L_\Sigma \sqrt{\bar{s}\chi_j^2}/\underline{s}^2$ and $\|\mathbf{A}^{-1}\boldsymbol{E}\| \le \kappa_j$. Under $\kappa_j < 1$, the quantitative implicit function theorem gives

$$\|\mathbf{a}_{j0}^* - \mathbf{a}_j^*\| \le \frac{\sqrt{d}\, \bar{s}\, L_\Sigma (\chi_j^2 + 1)}{2\underline{s}\, n_j^w (1 - \kappa_j)}. \tag{31}$$

Under the model, $w_{kj}^{(t)} \propto e^{-T_{kj}/2}$ with $T_{kj} = \chi_{k,j}^2$ having non-centrality $\Delta_{kj} = \sum_\ell n_k(\theta_{k,\ell} - a_{j,\ell}^*)^2 \sigma_{k,\ell}(\mathbf{a}_j^*)^{-2}$. For $\boldsymbol{\theta}_k \ne \mathbf{a}_j^*$, $\Delta_{kj} \to \infty$, so $w_{kj}^{(t)}/\alpha_{k_*,j} \to 0$ exponentially where k_* is the nearest client. Hence $\chi_j^2 = \sum_{k=1}^K w_{kj}^{(t)} T_{kj}$ is dominated by near clients contributing $\mathcal{O}_P(d)$ each, giving $\chi_j^2 = \mathcal{O}_P(d)$. The nearest client keeps a large enough weight, so $n_j^w \gtrsim \underline{n}$. Substituting into (31): $\kappa_j = o_P(1)$ and $\|\mathbf{a}_{j0}^* - \mathbf{a}_j^*\| = \mathcal{O}_P(d^{3/2}\bar{s}L_\Sigma/(\underline{s}\, n_j^w)) = o_P(1)$. □

Proof of Proposition A.1. For each $j \in [\hat{k}]$, let $C_j \in \mathcal{H}$ such that $\hat{\mathbf{a}}_j \in C_j$. Fix $\mathbf{u} \in \frac{1}{\sqrt{n_k}} \cdot C_j$ and $k \in [K]$, and let

$$\mathbf{z}_j(\mathbf{u}) := \Sigma_k(\mathbf{u})^{-1/2}(\underline{\hat{\boldsymbol{\theta}}}_k - \hat{\mathbf{a}}_j), \quad \mathbf{y}_j(\mathbf{u}) := \Sigma_k(\mathbf{u})^{-1/2}(\sqrt{n_k}\mathbf{u} - \hat{\mathbf{a}}_j).$$

Next, define a positive measure H_j supported on the corners of C_j such that $H_j(C_j) = \hat{w}_j$ and

$$\int_{C_j} (1 + \langle \mathbf{y}_j(\mathbf{u}), \mathbf{z}_j(\mathbf{u})\rangle) \, |2\pi\Sigma_k(\mathbf{u})|^{-1/2} \exp\left(-\frac{\|\mathbf{z}_j(\mathbf{u})\|^2}{2}\right) \mathrm{d}H_j(\mathbf{u})$$
$$\left\{= \hat{w}_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}; \hat{\mathbf{a}}_j)\right\} = \int_{C_j} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) \mathrm{d}\hat{G}^j(\mathbf{u})$$

for all $k \leq K$ where $\hat{G}^j := \hat{w}_j \delta_{\hat{\mathbf{a}}_j}$. By definitions $\|\mathbf{y}_j\| \leq \sqrt{\frac{d}{\underline{s}}}\delta$ and $\|\mathbf{z}_j\| \leq \sqrt{\frac{1}{\underline{s}}} D$ hence $\langle \mathbf{z}_j, \mathbf{y}_j\rangle^2 \leq 1$ by construction. Let $\phi_d(\cdot)$ denote the density of a $\mathcal{N}(0, I_d)$ distribution. First, we note that

$$\begin{aligned}
\varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) &= |\Sigma_k(\mathbf{u})|^{-1/2} \phi_d(\mathbf{z}_j - \mathbf{y}_j) \\
&= |\Sigma_k(\mathbf{u})|^{-1/2} \phi_d(\mathbf{z}_j) \exp(\langle \mathbf{z}_j, \mathbf{y}_j\rangle - \|\mathbf{y}_j\|^2/2)
\end{aligned}$$

and following Jiang and Zhang [2009, A.27]:

$$\begin{aligned}
(e^{-\|\mathbf{y}_j\|^2/2} - 1) e^{\langle \mathbf{z}_j, \mathbf{y}_j\rangle} &\leq e^{-\|\mathbf{y}_j\|^2/2 + \langle \mathbf{z}_j, \mathbf{y}_j\rangle} - (1 + \langle \mathbf{z}_j, \mathbf{y}_j\rangle) \\
&\leq \langle \mathbf{z}_j, \mathbf{y}_j\rangle^2 \underbrace{e^{\langle \mathbf{z}_j, \mathbf{y}_j\rangle - \|\mathbf{y}_j\|^2/2}}_{=:g}, \\
(\therefore) \quad g(1 - e^{\|\mathbf{y}_j\|^2/2}) &\leq g - (1 + \langle \mathbf{z}_j, \mathbf{z}_j\rangle) \leq \langle \mathbf{z}_j, \mathbf{y}_j\rangle^2 g
\end{aligned}$$

so we arrive at

$$\begin{aligned}
\varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u})(1 - e^{\|\mathbf{y}_j\|^2/2}) &\leq \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) - (1 + \langle \mathbf{z}_j, \mathbf{y}_j\rangle)|\Sigma_k(\mathbf{u})|^{-1/2} \phi_d(\mathbf{z}_j) \\
&\leq \langle \mathbf{z}_j, \mathbf{y}_j\rangle^2 \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) \leq \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}).
\end{aligned}$$

Hence, since $\int_{C_j} \varphi \, \mathrm{d}\hat{G}^j = \hat{w}_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \hat{\mathbf{a}}_j)$ is a single-atom evaluation while $\int_{C_j} \varphi \, \mathrm{d}H_j$ averages over corners, the sandwich above gives

$$\begin{aligned}
&\hat{w}_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \hat{\mathbf{a}}_j) - \int_{C_j} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) \mathrm{d}H_j(\mathbf{u}) \\
&\leq \frac{dD^2\delta^2}{\underline{s}^2} \hat{w}_j \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \hat{\mathbf{a}}_j) + \int_{C_j} \left(e^{\|\mathbf{y}_j\|^2/2} - 1\right) \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) H_j(\mathbf{u}) \\
&\leq \frac{dD^2\delta^2}{\underline{s}^2} \int_{C_j} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) \mathrm{d}\hat{G}^j(\mathbf{u}) + \left(e^{d\delta^2/(2\underline{s})} - 1\right) \int_{C_j} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \mathbf{u}) \mathrm{d}H_j(\mathbf{u}).
\end{aligned}$$

Let $H = \sum_{j=1}^{\hat{k}} H_j$ and let $\hat{G} = \sum_{j=1}^{\hat{k}} \hat{w}_j \delta_{\hat{\mathbf{a}}_j}$. Summing the above inequality over j,

$$f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k) - f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \leq \frac{dD^2\delta^2}{\underline{s}^2} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k) + \left(e^{\frac{d\delta^2}{2\underline{s}}} - 1\right) f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k), \text{ i.e.,}$$
$$f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)\left(1 - \frac{dD^2\delta^2}{\underline{s}^2}\right) \leq e^{d\delta^2/(2\underline{s})} f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k).$$

Since H is supported on $\mathcal{A}$, by optimality of $\hat{G}^{\mathcal{A}}$,

$$\prod_{k=1}^{K} f_{k,\hat{G}^{\mathcal{A}}}(\underline{\hat{\boldsymbol{\theta}}}_k) \geq \prod_{k=1}^{K} f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k).$$

Combining our findings,

$$\prod_{k=1}^{K} f_{k,\hat{G}^{\mathcal{A}}}(\underline{\hat{\boldsymbol{\theta}}}_k) \geq e^{-Kd\delta^2/(2\underline{s})} \prod_{k}\left(1 - \frac{dD^2\delta^2}{\underline{s}^2}\right) \prod_{k=1}^{K} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k).$$

Using the elementary inequality $1 - x \geq e^{-2x}$ for $x \leq 3/4$, we obtain

$$\prod_{k=1}^{K} f_{k,\hat{G}^{\mathcal{A}}}(\underline{\hat{\boldsymbol{\theta}}}_k) \geq \exp\left(-\frac{Kd\delta^2}{2\underline{s}} - \frac{2KdD^2\delta^2}{\underline{s}^2}\right) \prod_{k=1}^{K} f_{k,\hat{G}}(\underline{\hat{\boldsymbol{\theta}}}_k)$$

under the given upper limit for δ. □

Lemma D.1. *Fix a probability measure G on $\mathbb{R}^d$. We have*

$$\frac{1}{\bar{s}}\left(\frac{\|\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)\|}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)}\right)^2 \leq tr\left(\mathbf{I}_d + \frac{\nabla \mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}})}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)}\right) \leq \log \frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)^2} \tag{32}$$

where H stands for Hessian. Also for every $\hat{\boldsymbol{\theta}}_k \in \mathbb{R}^d$ and a measure G' on $\mathbb{R}^d$, we have

$$\frac{\|\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)\|}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \leq \sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d \rho^2}} \tag{33}$$

for $\rho \leq \frac{1}{\sqrt{e}(2\pi\underline{s})^{d/2}}$.

Proof. If $\boldsymbol{\theta} \sim G$ and $\underline{\hat{\boldsymbol{\theta}}}_k|\boldsymbol{\theta} \sim \mathcal{N}\left(\sqrt{n_k}\boldsymbol{\theta}, \Sigma_k(\boldsymbol{\theta})\right)$, then it follows that

$$\frac{\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)} = \mathbb{E}\left(\sqrt{n_k}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}_k | \underline{\hat{\boldsymbol{\theta}}}_k\right)$$

and

$$\frac{\nabla \mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)} = -\mathbf{I}_d + \mathbb{E}\left[(\sqrt{n_k}\boldsymbol{\theta} - \hat{\underline{\boldsymbol{\theta}}}_k)(\sqrt{n_k}\boldsymbol{\theta} - \hat{\underline{\boldsymbol{\theta}}}_k)'\Sigma_k(\boldsymbol{\theta})^{-1}|\hat{\underline{\boldsymbol{\theta}}}_k\right]. \tag{34}$$

It follows that

$$\operatorname{tr}\left(\mathbf{I}_d + \frac{\nabla \mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}\right) = \mathbb{E}[\|\Sigma_k(\boldsymbol{\theta})^{-1/2}(\sqrt{n_k}\boldsymbol{\theta} - \hat{\underline{\boldsymbol{\theta}}}_k)\|^2|\hat{\underline{\boldsymbol{\theta}}}_k] \geq \frac{\|\mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)\|^2}{\bar{s} f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}$$

and

$$\begin{aligned}\exp\left(\frac{1}{2}\mathbb{E}[\|\Sigma_k(\boldsymbol{\theta})^{-1/2}(\sqrt{n_k}\boldsymbol{\theta} - \hat{\underline{\boldsymbol{\theta}}}_k)\|^2|\hat{\underline{\boldsymbol{\theta}}}_k]\right) &\leq \mathbb{E}\left(e^{\frac{1}{2}\|\Sigma_k(\boldsymbol{\theta})^{-1/2}(\hat{\underline{\boldsymbol{\theta}}}_k - \sqrt{n_k}\boldsymbol{\theta})\|^2}|\hat{\underline{\boldsymbol{\theta}}}_k\right) \\ &= \int \frac{|2\pi\Sigma_k(\mathbf{u})|^{-1/2}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}\mathrm{d}G(\mathbf{u}) \leq \frac{(2\pi\underline{s})^{-d/2}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}\end{aligned}$$

so that we have

$$tr\left(\mathbf{I}_d + \frac{\nabla \mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}\right) \leq \log\frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}.$$

To prove (33), note first from (32) that

$$\begin{aligned}\frac{\|\mathbf{q}_{k,G}(\hat{\underline{\boldsymbol{\theta}}})\|}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)\vee\rho} &\leq \sqrt{\bar{s}\log\frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}}\frac{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)\vee\rho} \\ &= \begin{cases}\sqrt{\bar{s}\log\frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}} \leq \sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}} & \text{if } f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k) > \rho \\ \sqrt{\bar{s}\log\frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}}\frac{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{\rho} & \text{o/w.}\end{cases}\end{aligned}$$

The function $v \mapsto \sqrt{\log\frac{(2\pi\underline{s})^{-d}}{v^2}}v$ is non-decreasing on $(0, \frac{1}{e(2\pi\underline{s})^d}]$ hence when $f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2 \leq \rho^2 \leq \frac{1}{e(2\pi\underline{s})^d}$, the inequality

$$\sqrt{\bar{s}\log\frac{(2\pi\underline{s})^{-d}}{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)^2}}\frac{f_{k,G}(\hat{\underline{\boldsymbol{\theta}}}_k)}{\rho} \leq \sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}}$$

holds and this proves (33). □

The following lemma establishes a Lipschitz-type bound for the heteroskedastic Gaussian kernels that has no analogue in the fixed-covariance literature. Unlike the standard setting where the covariance is fixed, the gradient with respect to the parameter **u** now requires controlling the Jacobian of the variance function. This bound is the key new ingredient in our metric entropy estimates (Lemma E.3).

Lemma D.2. *Let* $\nabla_{\mathbf{u}}$ *denote* $\frac{\partial}{\partial \mathbf{u}}$ *and* $\mathbf{J} := \begin{bmatrix} \frac{\partial \sigma_{k,1}(\mathbf{u})}{\partial u_1} & \dots & \frac{\partial \sigma_{k,1}(\mathbf{u})}{\partial u_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial \sigma_{k,d}(\mathbf{u})}{\partial u_1} & \dots & \frac{\partial \sigma_{k,d}(\mathbf{u})}{\partial u_d} \end{bmatrix}$*. For any points* $\mathbf{x}, \mathbf{a}, \mathbf{b} \in \mathbb{R}^d$*, we have*

$$\left|\varphi^{(k)}(\mathbf{x};\mathbf{a}) - \varphi^{(k)}(\mathbf{x};\mathbf{b})\right| \leq_d \|\mathbf{a}-\mathbf{b}\| \frac{\sqrt{\bar{n}} + 2d^{3/2}\bar{s}'}{\underline{s}^{(d+1)/2}}$$

for $\bar{s}' := \|\mathbf{J}\|$.

Proof. Let ϕ_d denote the density of a d-dimensional standard normal distribution. We use mean value theorem: there exists some $\mathbf{u}$ which is some linear interpolation of $\mathbf{z_a} := \Sigma_k(\mathbf{a})^{-1/2}\left(\mathbf{x} - \sqrt{n_k}\mathbf{a}\right)$, $\mathbf{z_b}$ such that

$$\begin{aligned} &\frac{\phi_d(\mathbf{z_a})}{|\Sigma_k(\mathbf{a})|^{1/2}} - \frac{\phi_d(\mathbf{z_b})}{|\Sigma_k(\mathbf{b})|^{1/2}} = \nabla_{\mathbf{u}}\left(\frac{\phi_d(\mathbf{z_u})}{|\Sigma_k(\mathbf{u})|^{1/2}}\right)'(\mathbf{a}-\mathbf{b}) \\ \therefore\ &\left|\varphi^{(k)}(\mathbf{x};\mathbf{a}) - \varphi^{(k)}(\mathbf{x};\mathbf{b})\right| \leq \|\mathbf{a}-\mathbf{b}\| \sup_{\mathbf{u}\in\mathbb{R}^d} \nabla_{\mathbf{u}}\left(\frac{\phi_d(\mathbf{z_u})}{|\Sigma_k(\mathbf{u})|^{1/2}}\right). \end{aligned}$$

Now, observe that

$$\nabla_{\mathbf{u}}\left(\frac{\phi_d(\mathbf{z_u})}{|\Sigma_k(\mathbf{u})|^{1/2}}\right) = \frac{1}{|\Sigma_k(\mathbf{u})|^{\frac{1}{2}}}\left\{\nabla_{\mathbf{u}}\phi_d(\mathbf{z_u}) - \frac{d}{2}\phi_d(\mathbf{z_u})\nabla_{\mathbf{u}}\log|\Sigma_k(\mathbf{u})|\right\} \tag{35}$$

and using

$$\nabla_{\mathbf{u}}\mathbf{z_u} = -\sqrt{n_k}\Sigma_k(\mathbf{u})^{-1/2} - \Sigma_k(\mathbf{u})^{-1/2}\operatorname{diag}(\mathbf{z}_u)\begin{bmatrix} \nabla_{\mathbf{u}'}\sigma_{k,1}(\mathbf{u}) \\ \vdots \\ \nabla_{\mathbf{u}'}\sigma_{k,d}(\mathbf{u}) \end{bmatrix}_{d\times d},$$

we have

$$\|\nabla_{\mathbf{u}}\phi_d(\mathbf{z_u})\| = \frac{\phi_d(\mathbf{z_u})}{2}\left\|\nabla_{\mathbf{u}}\|\mathbf{z_u}\|^2\right\| \leq \frac{\phi_d(\mathbf{z_u})}{2}\frac{2\|\mathbf{z_u}\|}{\underline{s}^{1/2}}\left(\sqrt{n_k} + \|\mathbf{J}\|\cdot\|\mathbf{z_u}\|\right).$$

Also,

$$\|\nabla_{\mathbf{u}}\log|\Sigma_k(\mathbf{u})|\| = 2\left\|\mathbf{J}'\Sigma(\mathbf{u})^{-1/2}\mathbf{1}_d\right\| \leq \frac{2\sqrt{d}}{\underline{s}^{1/2}}\|\mathbf{J}\|$$

hence

$$\|(35)\| \leq \frac{\phi_d(\mathbf{z_u})}{\underline{s}^{\frac{d+1}{2}}} \left\{ \|\mathbf{z_u}\| \left(\sqrt{n_k} + \bar{s}'\|\mathbf{z_u}\|\right) + d^{3/2}\bar{s}' \right\} \leq_d \frac{1}{\underline{s}^{(d+1)/2}} \left(\frac{\sqrt{\bar{n}}}{\sqrt{e}} + \frac{2\bar{s}'}{e} + d^{3/2}\bar{s}' \right)$$
$$\leq \frac{\sqrt{\bar{n}} + 2d^{3/2}\bar{s}'}{\underline{s}^{(d+1)/2}}$$

uniformly over $\mathbf{u} \in \mathbb{R}^d$. □

Lemma D.3. *For a probability measure G on $\mathbb{R}^d$ and $\rho > 0$, let*

$$\tilde{\Delta}_k(G, \rho) := \int \left(1 - \frac{f_{k,G}}{f_{k,G} \vee \rho}\right)^2 \frac{\|\mathbf{q}_{k,G}\|^2}{f_{k,G}}.$$

Then,

1. *For every G and $\rho > 0$, we have $\tilde{\Delta}_k(G, \rho) \leq d\bar{s}$ for all $k \leq K$.*

2. *Suppose $G = \sum_{j=1}^m w_j H_j$ for some probability measures $H_1, \ldots, H_m$ and weights $w_1, \ldots, w_m$. Then*
$$\tilde{\Delta}_k(G, \rho) \leq \sum_{j=1}^m w_j \tilde{\Delta}_k\left(H_j, \rho/w_j\right). \tag{36}$$

Proof of Lemma D.3. Let us suppress the client index (k) for simplicity. Note that if $\boldsymbol{\theta} \sim G$ and $\underline{\hat{\boldsymbol{\theta}}}|\boldsymbol{\theta} \sim N\left(\sqrt{n}\boldsymbol{\theta}, \Sigma_k(\boldsymbol{\theta})\right)$, then

$$\frac{\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}})}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}})} = \mathbb{E}\left(\sqrt{n}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}|\underline{\hat{\boldsymbol{\theta}}}\right).$$

As a result

$$\tilde{\Delta}_k(G, \rho) \leq \int_{\mathbb{R}^d} \|\mathbb{E}\left(\sqrt{n}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}|\underline{\hat{\boldsymbol{\theta}}}\right)\|^2 f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}) \mathrm{d}\underline{\hat{\boldsymbol{\theta}}} = \mathbb{E}[\|\mathbb{E}(\sqrt{n}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}|\underline{\hat{\boldsymbol{\theta}}})\|^2]$$
$$\leq \mathbb{E}[\|\sqrt{n}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}\|^2] = \operatorname{tr}\mathbb{E}[\Sigma_k(\boldsymbol{\theta})].$$

The convexity inequality (36) is a direct consequence of the mixture decomposition in Saha and Guntuboyina [2020, Lemma F.7], which applies unchanged to our setting since it only involves the pointwise relationship between $f_{k,G}$ and $\mathbf{q}_{k,G}$. □

Lemma D.4. *Fix a probability measure G on $\mathbb{R}^d$ and let $0 < \rho \leq (2\pi\bar{s})^{-d/2}e^{-8/\underline{s}}$. Let $L(\rho) :=$*

$\sqrt{\underline{s}\log\frac{1}{(2\pi\bar{s})^d\rho^2}}$. *Then, for every compact set* $S\subseteq\mathbb{R}^d$, *we have*

$$\tilde{\Delta}_k\left(G,\rho\right)\le_{d,\underline{s},\bar{s}} N\left(\frac{4}{\sqrt{n_k}L(\rho)},S\right)\rho L(\rho)^d+G(S^c). \tag{37}$$

Proof of Lemma D.4. The bound follows by the same localization argument as Saha and Guntuboyina [2020, Lemma 4.3]: one decomposes $\tilde{\Delta}_k$ into contributions from $\{f_{k,G}\le\rho\}$ and its complement, controls the posterior mean on the former using the Gaussian tail, and covers the support of G with balls of radius $4/\{\sqrt{n_k}L(\rho)\}$. The heteroskedastic covariance enters only through the uniform eigenvalue bounds $\underline{s},\bar{s}$, so the calculation carries over with the same constants.

□

For the following lemma, we write $\partial_i^j f := \frac{\partial^j}{\partial x_i^j}f(\mathbf{x})$ to denote a partial derivative of a vector function $f:\mathbb{R}^d\to\mathbb{R}$.

Lemma D.5. *For every pair of probability measures* G *and* G_0 *on* $\mathbb{R}^d$, $1\le i\le d$, $j\ge 1$, *and* $k=1,...,K$, *we have*

$$\begin{aligned}&\int_{\mathbb{R}^d}\left\{\partial_i^j\left(\tilde{f}_{k,G,i}(\mathbf{x})-\tilde{f}_{k,G_0,i}(\mathbf{x})\right)\right\}^2\mathrm{d}\mathbf{x}\\&\quad\le\frac{\bar{s}}{(2\pi\underline{s})^{d/2}}\inf_{a\ge\sqrt{2j-1}}\left\{\sqrt{2\pi\underline{s}}a^{2j-1}e^{-a^2\underline{s}}+a^{2j}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right\}.\end{aligned} \tag{38}$$

Therefore, for a small enough Hellinger accuracy we have

$$\|\mathbf{q}_{k,G}-\mathbf{q}_{k,G_0}\|^2_{\mathcal{L}_2}\le_{d,\underline{s},\bar{s}}\sum_{i=1}^d h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\left|\log\sum_{i=1}^d h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right|$$

for $\|\boldsymbol{f}\|^2_{\mathcal{L}_2}:=\int\|\boldsymbol{f}(\mathbf{x})\|^2\mathrm{d}\mathbf{x}\ \forall\boldsymbol{f}:\mathbb{R}^d\to\mathbb{R}^d$.

The second conclusion can be derived for any non-negative, small enough $h(x)$ such that $1/h(x)\ge e^{2j-1}$ and $a=\max\left(\sqrt{2j-1},\sqrt{-\log h(x)}\right)$:

$$a^{2j-1}e^{-a^2}+a^{2j}h(x)\le 2a^{2j}h(x)\le 2h(x)\log^j 1/h(x).$$

Proof. The Plancherel-based approach adapts Saha and Guntuboyina [2020, Lemma F.2] to the variance-weighted densities $\tilde{f}_{k,G,i}$. The key modification is that Fourier transforms now act on $\sigma_{k,i}(\boldsymbol{\theta})^2$-weighted kernels, which introduces an additional factor of $\bar{s}$ in the tail bound; the algebraic structure of the iteration is otherwise unchanged. It suffices to prove the inequality for the $\tilde{f}_{i,\bullet}$ functions, as the score components follow by differentiation under the integral sign. Fix $a \geq \sqrt{2j-1}$ and assume, without loss of generality, that $i = 1$. Let

$$\bar{f}_{k,G,i'}(u) := \int e^{iux_1} \tilde{f}_{k,G,i'}(\mathbf{x}) \mathrm{d}x_1$$

denote the Fourier transform of $\tilde{f}_{k,G,i}$ treated as a univariate function. Plancherel's identity gives

$$\begin{aligned} 2\pi \int \left\{ \partial_1^j \left(\tilde{f}_{k,G,i}(\mathbf{x}) - \tilde{f}_{k,G_0,i}(\mathbf{x}) \right) \right\}^2 \mathrm{d}x_1 &= \int u^{2j} \left| \bar{f}_{k,G,i}(u) - \bar{f}_{k,G_0,i}(u) \right|^2 \mathrm{d}u \\ \leq a^{2j} \int \left| \bar{f}_{k,G,i}(u) - \bar{f}_{k,G_0,i}(u) \right|^2 \mathrm{d}u &+ \int_{|u|>a} u^{2j} \left| \bar{f}_{k,G,i}(u) - \bar{f}_{k,G_0,i}(u) \right|^2 \mathrm{d}u \\ = 2\pi a^{2j} \int \left(\tilde{f}_{k,G,i}(\mathbf{x}) - \tilde{f}_{k,G_0,i}(\mathbf{x}) \right)^2 \mathrm{d}x_1 &+ \int_{|u|>a} u^{2j} \left| \bar{f}_{k,G,i}(u) - \bar{f}_{k,G_0,i}(u) \right|^2 \mathrm{d}u \end{aligned} \tag{39}$$

for every $a > 0$. Also note that for every $u, x_2, \ldots, x_d \in \mathbb{R}$,

$$\begin{aligned} \bar{f}_{k,G,i}(u) &= \iint e^{iux_1} \sigma_{k,i}(\boldsymbol{\theta})^2 |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} \exp\left\{ -\sum_{j=1}^{d} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2} \right\} \mathrm{d}x_1 \mathrm{d}G(\boldsymbol{\theta}) \\ &\leq \int \mathbf{x}s \frac{\bar{s}\sqrt{2\pi\sigma_{k,1}(\boldsymbol{\theta})^2}}{|2\pi\Sigma_k(\boldsymbol{\theta})|^{1/2}} e^{-\sum_{j\neq 1} \frac{(x_j-\sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2}} \int \frac{e^{iux_1}}{\sqrt{2\pi\sigma_{k,1}(\boldsymbol{\theta})^2}} e^{-\frac{(x_1-\sqrt{n_k}\theta_1)^2}{2\sigma_{k,1}(\boldsymbol{\theta})^2}} \mathrm{d}x_1 \mathrm{d}G(\boldsymbol{\theta}) \\ &= \int \frac{\bar{s}(2\pi)^{-(d-1)/2}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})} \exp\left(iu\sqrt{n_k}\theta_1 - \frac{u^2\sigma_{k,1}(\boldsymbol{\theta})^2}{2} \right) \exp\left\{ -\sum_{j\neq 1} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2} \right\} \mathrm{d}G(\boldsymbol{\theta}) \end{aligned}$$

so that

$$\left| \bar{f}_{k,G,i}(u) \right| \leq e^{-u^2\underline{s}/2} \int \frac{(2\pi)^{-(d-1)/2}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})} \exp\left\{ -\sum_{j\neq 1} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2} \right\} \mathrm{d}G(\boldsymbol{\theta}).$$

Similarly, we have

$$\begin{aligned} \bar{f}_{k,G}(u) &= \iint e^{iux_1} |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} \exp\left\{ -\sum_{j=1}^{d} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2} \right\} \mathrm{d}x_1 \mathrm{d}G(\boldsymbol{\theta}) \\ &\leq \int \mathbf{x}s \frac{\sqrt{2\pi\sigma_{k,1}(\boldsymbol{\theta})^2}}{|2\pi\Sigma_k(\boldsymbol{\theta})|^{1/2}} e^{-\sum_{j\neq 1} \frac{(x_j-\sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2}} \int \frac{e^{iux_1}}{\sqrt{2\pi\sigma_{k,1}(\boldsymbol{\theta})^2}} e^{-\frac{(x_1-\sqrt{n_k}\theta_1)^2}{2\sigma_{k,1}(\boldsymbol{\theta})^2}} \mathrm{d}x_1 \mathrm{d}G(\boldsymbol{\theta}) \\ &= \int \frac{(2\pi)^{-(d-1)/2}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})} \exp\left(iu\sqrt{n_k}\theta_1 - \frac{u^2\sigma_{k,1}(\boldsymbol{\theta})^2}{2} \right) \exp\left\{ -\sum_{j\neq 1} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{2\sigma_{k,j}(\boldsymbol{\theta})^2} \right\} \mathrm{d}G(\boldsymbol{\theta}). \end{aligned}$$

So, the second term in (39) can be bounded from above as

$$\int_{|u|>a} u^{2j} \left|\bar{f}_{k,G,i}(u) - \bar{f}_{k,G_0,i}(u)\right|^2 \mathrm{d}u$$
$$\leq 2 \int \frac{\bar{s}(2\pi)^{-(d-1)}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})^2} \exp\left\{-\sum_{j\neq 1} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{\sigma_{k,j}(\boldsymbol{\theta})^2}\right\} \mathrm{d}(G+G_0)(\boldsymbol{\theta}) \int_{|u|>a} u^{2j} e^{-u^2\underline{s}}\mathrm{d}u.$$

Noting that

$$\begin{aligned}\iint \frac{2\bar{s}(2\pi)^{-(d-1)}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})^2} \exp\left\{-\sum_{j\neq 1} \frac{(x_j - \sqrt{n_k}\theta_j)^2}{\sigma_{k,j}(\boldsymbol{\theta})^2}\right\} \mathrm{d}\mathbf{x}_{-1}\mathrm{d}(G+G_0)(\boldsymbol{\theta}) \\ = \int \frac{2\bar{s}(2\pi)^{(1-d)/2}}{\prod_{j\neq 1} \sigma_{k,j}(\boldsymbol{\theta})}\mathrm{d}(G+G_0)(\boldsymbol{\theta}) \leq \frac{4\bar{s}(2\pi)^{(1-d)/2}}{\underline{s}^{(d-1)/2}},\end{aligned}$$

we deduce that

$$\begin{aligned}\int \left\{\partial_1^j \left(\tilde{f}_{k,G,i}(\mathbf{x}) - \tilde{f}_{k,G_0,i}(\mathbf{x})\right)\right\}^2 \mathrm{d}\mathbf{x} \leq 2\pi a^{2j} \int \left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)^2 \\ + 4\bar{s}(2\pi/\underline{s})^{(d-1)/2} \int_{|u|>a} u^{2j} e^{-u^2\underline{s}}\mathrm{d}u.\end{aligned}$$

Note that

$$\int_{|u|>a} u^{2j} e^{-u^2\underline{s}}\mathrm{d}u \leq a^{2j-1}e^{-a^2\underline{s}}$$

for $a \geq \sqrt{2j-1}$ by integration by parts and deduction. Also, by $(a-b)^2 = (\sqrt{a}-\sqrt{b})^2(\sqrt{a}+\sqrt{b})^2$,

$$\int \left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)^2 \leq \frac{\bar{s}}{(2\pi\underline{s})^{d/2}} h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)$$

as $\tilde{f}_{k,G}$ is bounded above by $\bar{s}\,(2\pi\underline{s})^{-d/2}$ for any G. $\square$

Theorem D.6. *For every pair of probability measures G and G_0 on $\mathbb{R}^d$ and $0 < \rho \leq \sqrt{\frac{1}{(2\pi\bar{s})^d e}}$, the quantity*

$$\mathbb{E}_{\underline{\hat{\boldsymbol{\theta}}}_k \sim f_{k,G_0}} \left\| \frac{\mathbf{q}_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G_0}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} - \frac{\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k)}{f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) \vee \rho} \right\|^2 \tag{40}$$

is bounded above by

$$\sum_{i=1}^{d} \max\left\{ \left(\log \frac{(2\pi \underline{s})^{-d}}{\rho^2}\right)^3, \left|\log h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)\right| \right\} h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right) \\ + h^2(f_{k,G}, f_{k,G_0}) \log \frac{(2\pi \underline{s})^{-d}}{\rho^2} \qquad (41)$$

up to the multiplication by a constant that depends on $\underline{s}, \bar{s}$ *only.*

Proof. For functions $\mathbf{u} : \mathbb{R}^d \to \mathbb{R}^d$, we let

$$\|\mathbf{u}\|_{0,k} := \left(\int \|\mathbf{u}(\mathbf{x})\|^2 f_{k,G_0}(\mathbf{x}) \mathrm{d}\mathbf{x}\right)^{1/2}.$$

Set

$$m_k := f_{k,G} + f_{k,G_0}, \quad m_{\rho,k} := f_{k,G} \vee \rho + f_{k,G_0} \vee \rho$$

so that,

$$\begin{aligned}
&\left\| \frac{\mathbf{q}_{k,G_0}}{f_{k,G_0} \vee \rho} - \frac{\mathbf{q}_{k,G}}{f_{k,G} \vee \rho} \right\|_{0,k} \\
&= \left\| \frac{\mathbf{q}_{k,G_0}}{f_{k,G_0} \vee \rho} + \frac{-2\mathbf{q}_{k,G_0} + 2\mathbf{q}_{k,G_0} - 2\mathbf{q}_{k,G} + 2\mathbf{q}_{k,G}}{m_{\rho,k}} - \frac{\mathbf{q}_{k,G}}{f_{k,G} \vee \rho} \right\|_{0,k} \\
&\leq 2 \max_{H \in \{G, G_0\}} \left\| \frac{(\mathbf{q}_{k,H}) |f_{k,G} \vee \rho - f_{k,G_0} \vee \rho|}{(f_{k,H} \vee \rho) m_{\rho,k}} \right\|_{0,k} + 2 \left\| \frac{\mathbf{q}_{k,G} - \mathbf{q}_{k,G_0}}{m_{\rho,k}} \right\|_{0,k}
\end{aligned}$$

which is the standard T_1/T_2 decomposition of Saha and Guntuboyina [2020, Theorem E.1], applied here to the variance-weighted score $\mathbf{q}_{k,G}$. Write the RHS above by T_1 and T_2 respectively so that we have

$$(40) \lesssim \mathbb{E}(T_1^2) + \mathbb{E}(T_2^2).$$

We shall now bound T_1 and T_2 separately.

Bound on T_1^2. For T_1, we use inequality (33) in Lemma D.1 (note that $\rho \leq \sqrt{\frac{1}{(2\pi\bar{s})^d e}}$). That is,

$$
\begin{aligned}
\mathbb{E}(T_1^2) &= \max_{H\in\{G,G_0\}} \int \frac{\|\mathbf{q}_{k,H}\|^2 (f_{k,G}\vee\rho - f_{k,G_0}\vee\rho)^2}{(f_{k,H}\vee\rho)^2\, m_{\rho,k}^2} f_{k,G_0} \\
&\leq \bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2} \int \frac{(f_{k,G}-f_{k,G_0})^2}{m_{\rho,k}^2} f_{k,G_0} \\
&= \bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2} \int \left(\sqrt{f_{k,G}}-\sqrt{f_{k,G_0}}\right)^2 \frac{\left(\sqrt{f_{k,G}}+\sqrt{f_{k,G_0}}\right)^2}{m_{\rho,k}^2} f_{k,G_0} \\
&\leq \bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2} \int \left(\sqrt{f_{k,G}}-\sqrt{f_{k,G_0}}\right)^2 \frac{2m_k f_{k,G_0}}{m_{\rho,k}^2} \\
&\leq \bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2} \int \left(\sqrt{f_{k,G}}-\sqrt{f_{k,G_0}}\right)^2 = \bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}\cdot h^2\left(f_{k,G},f_{k,G_0}\right).
\end{aligned}
$$

Bound on T_2^2. Now, recalling that $\mathbf{q}_{k,G}(\cdot) = \begin{bmatrix} \partial_1\tilde{f}_{k,G}(\cdot) \\ \vdots \\ \partial_d\tilde{f}_{k,G}(\cdot)\end{bmatrix}$, start by writing

$$
\begin{aligned}
\mathbb{E}(T_2^2) = \int \frac{\|\mathbf{q}_{k,G}-\mathbf{q}_{k,G_0}\|^2}{m_{\rho,k}^2} f_{k,G_0} &\leq \int \frac{\|\mathbf{q}_{k,G}-\mathbf{q}_{k,G_0}\|^2}{m_{\rho,k}} \\
&= \sum_{i=1}^d \int \frac{\left\{\partial_i\left(\tilde{f}_{k,G,i}-\tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}} =: \sum_{i=1}^d D_{i,1}^2
\end{aligned}
\tag{42}
$$

where, for $1\leq i\leq d$ and $j\geq 0$,

$$
D_{i,j}^2 := \int \frac{\left\{\partial_i^j\left(\tilde{f}_{k,G,i}-\tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}} \quad\text{with}\quad \partial_i^j := \frac{\partial^j}{\partial(\cdot)_i^j}.
$$

We bound $D_{i,1}^2$ via a recursive inequality relating consecutive orders $D_{i,j-1}, D_{i,j}, D_{i,j+1}$ (cf. the iteration in Saha and Guntuboyina [2020, Theorem E.1], applied here to $\tilde{f}_{k,G,i}$). First, we acknowledge that

$$
\begin{aligned}
D_{i,0}^2 = \int \frac{\left(\tilde{f}_{k,G,i}-\tilde{f}_{k,G_0,i}\right)^2}{m_{\rho,k}} &\leq 2\int \left(\sqrt{\tilde{f}_{k,G,i}}-\sqrt{\tilde{f}_{k,G_0,i}}\right)^2 \frac{\tilde{f}_{k,G,i}+\tilde{f}_{k,G_0,i}}{m_{\rho,k}} \\
&\leq 2\int \left(\sqrt{\tilde{f}_{k,G,i}}-\sqrt{\tilde{f}_{k,G_0,i}}\right)^2 \frac{\bar{s}m_k}{m_{\rho,k}} \\
&\leq 2\bar{s}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right).
\end{aligned}
\tag{43}
$$

Integration by parts gives, for $j \geq 1$,

$$
\begin{aligned}
D_{i,j}^2 = &-\int \left[\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right]\left[\partial_i^{j}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right]\partial_i\left(\frac{1}{m_{\rho,k}}\right) \\
&- \int \frac{\left[\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right]\left[\partial_i^{j+1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right]}{m_{\rho,k}}.
\end{aligned} \tag{44}
$$

Almost surely, we have

$$
\begin{aligned}
\left|\partial_i\left(\frac{1}{m_{\rho,k}}\right)\right| &\leq \frac{|\partial_i f_{k,G}| + |\partial_i f_{k,G_0}|}{m_{\rho,k}^2} \leq \frac{|\partial_i f_{k,G}|/f_{k,G} \vee \rho + |\partial_i f_{k,G_0}|/f_{k,G_0} \vee \rho}{m_{\rho,k}} \\
&\leq \frac{\|\nabla f_{k,G}\|/f_{k,G} \vee \rho + \|\nabla f_{k,G_0}\|/f_{k,G_0} \vee \rho}{m_{\rho,k}} \leq \frac{2}{m_{\rho,k}}\sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}}
\end{aligned}
$$

by applying (33) in Lemma D.1. Therefore,

$$
\begin{aligned}
D_{i,j}^2 \leq 2\sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}} &\int \frac{\left|\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right|\left|\partial_i^{j}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right|}{m_{\rho,k}} \\
&+ \int \frac{\left|\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right|\left|\partial_i^{j+1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right|}{m_{\rho,k}}.
\end{aligned}
$$

Applying the Cauchy-Schwarz inequality to each of the two terms on the RHS above, we obtain

$$
\begin{aligned}
D_{i,j}^2 \leq 2\sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}} &\sqrt{\int \frac{\left\{\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}}}\sqrt{\int \frac{\left\{\partial_i^{j}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}}} \\
+ &\sqrt{\int \frac{\left\{\partial_i^{j-1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}}}\sqrt{\int \frac{\left\{\partial_i^{j+1}\left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right\}^2}{m_{\rho,k}}}
\end{aligned}
$$

which can be rewritten as

$$
\frac{D_{i,j}}{D_{i,j-1}} \leq \Upsilon + \frac{D_{i,j+1}}{D_{i,j}}, \quad j \geq 1 \tag{45}
$$

where $\Upsilon := 2\sqrt{\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}}$. Fix an integer $j_0 \geq 1$ and a real number $\beta > 0$.

Case 1. Suppose first that there exists an integer $1 \leq j \leq j_0$ such that $D_{i,j+1} \leq \beta D_{i,j}$. Then applying (45) recursively for $1, \ldots, j$, we obtain

$$
\frac{D_{i,1}}{D_{i,0}} \leq j\Upsilon + \beta
$$

so that, by (43),

$$D_{i,1} \leq \sqrt{2\bar{s}}\,(j_0\Upsilon + \beta)\, h\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right). \tag{46}$$

Case 2. Now suppose that $D_{i,j+1} > \beta D_{i,j}$ for every integer $1 \leq j \leq j_0$. In this case, we deduce from (45) that

$$\frac{D_{i,j}}{D_{i,j-1}} \leq \left(1 + \frac{\Upsilon}{\beta}\right) \frac{D_{i,j+1}}{D_{i,j}}, \quad j \leq j_0.$$

A recursive application of this inequality implies that

$$\frac{D_{i,1}}{D_{i,0}} \leq \left(1 + \frac{\Upsilon}{\beta}\right)^j \frac{D_{i,j+1}}{D_{i,j}}, \quad j \leq j_0.$$

Taking the geometric mean of the above inequality for $j = 0, 1, \ldots, j_0$, we have

$$D_{i,1} \leq \left(1 + \frac{\Upsilon}{\beta}\right)^{j_0/2} D_{i,j_0+1}{}^{\frac{1}{j_0+1}} D_{i,0}{}^{\frac{j_0}{j_0+1}}.$$

From Lemma D.5, we deduce that

$$\begin{aligned} D_{i,j}^2 &\leq \frac{1}{\rho} \int \left\{\partial_i^j \left(\tilde{f}_{k,G,i} - \tilde{f}_{k,G_0,i}\right)\right\}^2 \\ &\leq_d \frac{\bar{s}}{\rho(2\pi\underline{s})^{d/2}} \left\{\sqrt{2\pi\underline{s}}\, a^{2j-1} e^{-a^2\underline{s}} + a^{2j} h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)\right\} \end{aligned} \tag{47}$$

for $a \geq \sqrt{2j-1}$. Now using (43) and the bound (47) (with $j = j_0 + 1$), we obtain

$$\begin{aligned} D_{i,1} \leq \left(1 + \frac{\Upsilon}{\beta}\right)^{\frac{j_0}{2}} &\left[\frac{\bar{s}}{\rho(2\pi\underline{s})^{d/2}} \left\{\sqrt{2\pi\underline{s}}\, a^{2j_0+1} e^{-a^2\underline{s}} + a^{2(j_0+1)} h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)\right\}\right]^{\frac{1/2}{j_0+1}} \\ &\times \left\{2\bar{s} h^2\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)\right\}^{\frac{j_0/2}{j_0+1}} \end{aligned} \tag{48}$$

for every $a \geq \sqrt{2j_0+1}$.

The final bound obtained for $D_{i,1}$ is the maximum of cases 1 and 2. β will be chosen as $\beta = j_0\Upsilon$ so that

$$(46) \leq 3\sqrt{\bar{s}}\, j_0 \Upsilon h\left(\tilde{f}_{k,G,i}, \tilde{f}_{k,G_0,i}\right)$$

and

$$(48) \le \sqrt{e}\left[\frac{a^{2j_0+1}}{\rho/\bar{s}}\left(\frac{\tau}{2\pi}\right)^{d/2}\left\{ah^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)+\sqrt{2\pi\underline{s}}e^{-a^2\underline{s}}\right\}\right]^{\frac{1/2}{j_0+1}}\left\{\bar{s}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right\}^{\frac{j_0/2}{j_0+1}}.$$

Noting that

$$\left(\frac{1}{(2\pi\underline{s})^{d/2}\rho}\right)^{\frac{1/2}{j_0+1}} = \exp\left\{\Upsilon^2/(2\bar{s})\right\}^{\frac{1/2}{j_0+1}},$$

take j_0 to be the smallest integer ≥ 1 such that $j_0+1 \ge \frac{\Upsilon^2}{2\bar{s}}$. Then, the above term is at most $\sqrt{e}$. Finally, a will be taken to be

$$a := \max\left\{\sqrt{2j_0+1}, \sqrt{\frac{-1}{\underline{s}}\log h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)}, \sqrt{2\pi\underline{s}}\right\}$$

so that $e^{-a^2\underline{s}} \le h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)$ hence (48) can then be further bounded by

$$\begin{aligned}
&ea^{\frac{j_0+1/2}{j_0+1}}\left\{a\bar{s}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)+\bar{s}\sqrt{2\pi\underline{s}}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right\}^{\frac{1/2}{j_0+1}}\left\{\bar{s}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right\}^{\frac{j_0/2}{j_0+1}}\\
&\le ea^{\frac{j_0+1/2}{j_0+1}}\left(a+\sqrt{2\pi\underline{s}}\right)^{\frac{1/2}{j_0+1}}\sqrt{\bar{s}}h\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right) \le ea\sqrt{\bar{s}}h\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right).
\end{aligned}$$

Because $a \ge 1$ and $D_{i,1}$ is bounded by the maximum of the bounds given by (46) and (48), we obtain:

$$D_{i,1} \lesssim \sqrt{\bar{s}}\max\left(j_0\Upsilon, a\right)h\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right).$$

Our choice of j_0 also satisfies $j_0 \le \frac{\Upsilon^2}{2\bar{s}}$ so $j_0\Upsilon \le \Upsilon^3/\bar{s} = \frac{8}{\bar{s}}\left(\bar{s}\log\frac{1}{(2\pi\underline{s})^d\rho^2}\right)^{3/2}$ gives

$$D_{i,1} \le_{\bar{s},\underline{s}} \max\left\{\left(\log\frac{1}{(2\pi\underline{s})^d\rho^2}\right)^{\frac{3}{2}}, \sqrt{\frac{1}{\underline{s}}\left|\log h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right|}\right\}h\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right).$$

Combining with (42), we deduce that

$$E(T_2^2) \le_{d,\bar{s},\underline{s}} \sum_{i=1}^{d}\max\left\{\log^3\frac{(2\pi\underline{s})^{-d}}{\rho^2}, \left|\log h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right)\right|\right\}h^2\left(\tilde{f}_{k,G,i},\tilde{f}_{k,G_0,i}\right).$$

Combine the above finding with the bound for T_1^2 to finish the proof. □

The following lemma extends the tail moment bound of Soloff et al. [2025, Lemma 15] to our heteroskedastic kernels.

Lemma D.7. *For any $\lambda \in (0, 1 \wedge q]$ and $M > 1$,*

$$\mathbb{E}\left[\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)^\lambda 1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M}\right] \leq 2C_d M^{d+\lambda-2}\bar{s}^{1-d/2}e^{-M^2/(8\bar{s})} + M^\lambda \left(\frac{2\mu_{k,q}(S, G_0)}{M}\right)^q$$

where $\mu_{k,q}(S, G_0) := \mathbb{E}_{\boldsymbol{\theta}\sim G_0}\left[\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta})^q\right]^{1/q}$.

Proof. Note that $\underline{\hat{\boldsymbol{\theta}}}_k = \sqrt{n_k}\boldsymbol{\theta}_k + \Sigma_k(\boldsymbol{\theta}_k)^{1/2}\mathbf{z}_k$ where $\mathbf{z}_k \sim \mathcal{N}_d(0, \mathbf{I}_d)$. The proof adapts Soloff et al. [2025, Lemma 15]; the only modification is that $\Sigma_k(\boldsymbol{\theta}_k)^{1/2}$ replaces the fixed scale matrix, which is absorbed by the uniform bound $\bar{s}$. Since distance $\mathfrak{d}_S$ is 1-Lipschitz,

$$\begin{aligned}
&\mathbb{E}\left[\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)^\lambda 1_{\mathfrak{d}_S(\underline{\hat{\boldsymbol{\theta}}}_k)\geq M}\right] \\
&\quad\leq \mathbb{E}\left[(2\|\Sigma_k(\boldsymbol{\theta}_k)^{1/2}\mathbf{z}_k\|)^\lambda 1_{2\|\Sigma_k(\boldsymbol{\theta}_k)^{1/2}\mathbf{z}_k\|\geq M}\right] + \mathbb{E}\left[(2\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta}_k))^\lambda 1_{2\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta}_k)\geq M}\right].
\end{aligned}$$

The first term is bounded above by

$$\begin{aligned}
&2M^{\lambda-1}\mathbb{E}\left[\left\|\Sigma_k(\boldsymbol{\theta}_k)^{1/2}\mathbf{z}_k\right\| 1_{\|\Sigma_k(\boldsymbol{\theta}_k)^{1/2}\mathbf{z}_k\|\geq M/2}\right] \\
&\qquad\leq 2M^{\lambda-1}\sqrt{\bar{s}}\mathbb{E}\left[\|\mathbf{z}_k\|1_{\left\{\|\mathbf{z}_k\|\geq \frac{M}{2\bar{s}^{1/2}}\right\}}\right] \leq 2C_d M^{\lambda-1}\sqrt{\bar{s}}\left(\frac{M}{\bar{s}^{1/2}}\right)^{d-1} e^{-M^2/(8\bar{s})}.
\end{aligned}$$

Since $\lambda < q$, applying Hölder yields

$$\mathbb{E}\left[(2\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta}_k))^\lambda 1_{2\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta}_k)\geq M}\right] \leq M^\lambda\left(\frac{2\mu_{k,q}(S, G_0)}{M}\right)^q$$

where $\mu_{k,q}(S, G_0)$ denotes the q-th moment of $\mathfrak{d}_S(\sqrt{n_k}\boldsymbol{\theta}_k)$. ☐

E Control of Metric Entropy

To bound the metric entropy of the induced marginal classes, we adopt the moment-matching techniques.

Lemma E.1. *(Moment matching, part i) Let $a > 0$ and $G, H \in \mathcal{P}(\mathbb{R}^d)$. Suppose $A \subset \mathbb{R}^d$ is such that*

$$\mathbb{B}_a(\hat{\boldsymbol{\theta}}_k) \subseteq A \subseteq \mathbb{B}_{ca}(\hat{\boldsymbol{\theta}}_k), \quad c \geq 1$$

for all $k \leq K$ and

$$\int_A \frac{\prod_{l=1}^d \theta_l^{\alpha_l}}{\prod_{j=1}^d \sigma_{k,j}(\boldsymbol{\theta})^{2\beta_j+1}} \mathrm{d}(G-H)(\boldsymbol{\theta}) = 0, \quad k = 1, \ldots, K \tag{49}$$

for all multi-indices $\alpha, \beta \in \mathbb{N}_{\geq 0}^d$ such that $\|\alpha\|_1 \leq 2m+1, \|\beta\|_1 \leq m$. Then,

$$\max_{1 \leq k \leq K} \left| f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \right| \leq \frac{2}{(2\pi \underline{s})^{d/2}} \left\{ \left(\frac{ec^2 \bar{n} a^2}{2\underline{s}(m+1)} \right)^{m+1} + e^{-\underline{n}a^2/(2\bar{s})} \right\}$$

and

$$\max_{1 \leq k \leq K} \left\| \mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - \mathbf{q}_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \right\| \leq \frac{2\sqrt{\bar{n}}a}{(2\pi\underline{s})^{d/2}} \left\{ c \left(\frac{ec^2 \bar{n} a^2}{2\underline{s}(m+1)} \right)^{m+1} + \sqrt{\frac{\bar{s}}{e\bar{n}a^2}} + e^{-\underline{n}a^2/(2\bar{s})} \right\}.$$

Proof. For each $k \in [K]$, we have

$$f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) = \int_{A \cup A^c} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \mathrm{d}(G-H)(\boldsymbol{\theta}).$$

When $\boldsymbol{\theta} \in A^c$, $\|\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}\| \geq a$ hence $\|\underline{\hat{\boldsymbol{\theta}}}_k - \sqrt{n_k}\boldsymbol{\theta}_k\| \geq \sqrt{n_k}a$, so

$$\int_{A^c} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \mathrm{d}(G-H)(\boldsymbol{\theta}) \leq \frac{2e^{-n_k a^2/(2\bar{s})}}{(2\pi\underline{s})^{d/2}}.$$

Next, write $\mathbf{z}_{k,\boldsymbol{\theta}} := \Sigma_k(\boldsymbol{\theta})^{-1/2}\left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\right)$. The remainder term of the polynomial expansion of

$$|2\pi\Sigma_k(\boldsymbol{\theta})|^{1/2} \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) = \exp\left(-n_k \|\mathbf{z}_{k,\boldsymbol{\theta}}\|^2/2\right)$$

with degree m, $P(\mathbf{z}_{k,\boldsymbol{\theta}}) = \sum_{j \leq m} \frac{(-n_k\|\mathbf{z}_{k,\boldsymbol{\theta}}\|^2)^j}{2^j j!}$, satisfies [Saha and Guntuboyina, 2020, Lemma

D.2]:

$$|R(\mathbf{z}_{k,\boldsymbol{\theta}})| \leq \frac{(n_k \|\mathbf{z}_{k,\boldsymbol{\theta}}\|^2/2)^{m+1}}{(m+1)!} \leq \left\{ \frac{en_k \|\mathbf{z}_{k,\boldsymbol{\theta}}\|^2}{2(m+1)} \right\}^{m+1} \leq \left\{ \frac{en_k \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2}{2\underline{s}(m+1)} \right\}^{m+1}.$$

Now,

$$\varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) = |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} \{P(\mathbf{z}_{k,\boldsymbol{\theta}}) + R(\mathbf{z}_{k,\boldsymbol{\theta}})\}$$

and since $\int_A |\Sigma_k(\boldsymbol{\theta})|^{-1/2} P(\mathbf{z}_{k,\boldsymbol{\theta}}) \,\mathrm{d}(G-H)(\boldsymbol{\theta}) = \int_A |\Sigma_k(\boldsymbol{\theta})|^{-1/2} \theta_i P(\mathbf{z}_{k,\boldsymbol{\theta}}) \,\mathrm{d}(G-H)(\boldsymbol{\theta}) = 0$ for $i = 1, ..., d$,

$$\begin{aligned}
\left| \int_A \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \mathrm{d}(G-H)(\boldsymbol{\theta}) \right| &\leq \left| \int_A |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} R(\mathbf{z}_{k,\boldsymbol{\theta}}) \mathrm{d}(G-H)(\boldsymbol{\theta}) \right| \\
&\leq \frac{2}{(2\pi\underline{s})^{d/2}} \left\{ \frac{en_k c^2 a^2}{2\underline{s}(m+1)} \right\}^{m+1}.
\end{aligned}$$

Next, recall that $\mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) = \int (\sqrt{n_k}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}_k) \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \mathrm{d}G(\boldsymbol{\theta})$. We have

$$\int_{A^c} \|\sqrt{n_k}\boldsymbol{\theta} - \underline{\hat{\boldsymbol{\theta}}}_k\| \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) |\mathrm{d}(G-H)(\boldsymbol{\theta})| \leq \sup_{u \geq \sqrt{n_k} a} \frac{2ue^{-u^2/(2\bar{s})}}{(2\pi\underline{s})^{d/2}}.$$

Case-by-case analysis of the last term depends on whether the starting point $\sqrt{n_k}a$ lies before or after the critical point $\sqrt{\bar{s}}$. In short, we have

$$\sup_{u \geq \sqrt{n_k} a} \frac{ue^{-u^2/(2\bar{s})}}{(2\pi\underline{s})^{d/2}} = \frac{\sqrt{\bar{s}/e}}{(2\pi\underline{s})^{d/2}} 1_{\left\{a \leq \sqrt{\bar{s}/n_k}\right\}} + \frac{\sqrt{n_k} a e^{-n_k a^2/(2\bar{s})}}{(2\pi\underline{s})^{d/2}} 1_{\left\{a > \sqrt{\bar{s}/n_k}\right\}}.$$

Finally,

$$\begin{aligned}
&\left\| \int_A |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} (\underline{\hat{\boldsymbol{\theta}}}_k - \sqrt{n_k}\boldsymbol{\theta}) \varphi^{(k)}(\underline{\hat{\boldsymbol{\theta}}}_k; \boldsymbol{\theta}) \mathrm{d}(G-H)(\boldsymbol{\theta}) \right\| \\
&\leq \left\| \int_A |2\pi\Sigma_k(\boldsymbol{\theta})|^{-1/2} (\underline{\hat{\boldsymbol{\theta}}}_k - \sqrt{n_k}\boldsymbol{\theta}) R(\mathbf{z}_{k,\boldsymbol{\theta}}) \mathrm{d}(G-H)(\boldsymbol{\theta}) \right\| \\
&\leq \frac{2\sqrt{n_k} ca}{(2\pi\underline{s})^{d/2}} \left\{ \frac{en_k c^2 a^2}{2\underline{s}(m+1)} \right\}^{m+1}.
\end{aligned}$$

□

Lemma E.2. *(Moment matching, part ii) For any $G \in \mathcal{P}(\mathbb{R}^d)$, there is a discrete distribution H supported on S^a with at most*

$$l := (\lfloor 27\bar{n}a^2/\underline{s} \rfloor + 2)^d N(a, S^a) + 1, \quad a > 0$$

atoms such that

$$\|f_{\bullet,G} - f_{\bullet,H}\|_{\infty,S^a} \leq \frac{4e^{-\underline{n}a^2/(2\bar{s})}}{(2\pi\underline{s})^{d/2}}, \quad \|\mathbf{q}_{\bullet,G} - \mathbf{q}_{\bullet,H}\|_{\infty,S^a} \leq \frac{6a}{(2\pi\underline{s})^{d/2}} \left\{ \sqrt{\frac{\bar{s}}{e}} + 2\sqrt{\bar{n}}e^{-\underline{n}a^2/(2\bar{s})} \right\}.$$

Proof. The covering argument follows Saha and Guntuboyina [2020, Lemma D.3]; the moment-matching condition (49) now involves the variance functions $\sigma_{k,j}(\boldsymbol{\theta})$, and the remainder bounds must absorb the heteroskedastic kernel via the uniform eigenvalue bounds. Let $\mathring{S}^a := \cup_{\hat{\boldsymbol{\theta}} \in S} \mathbb{B}_a^o(\hat{\boldsymbol{\theta}})$ (here $\mathbb{B}_a^o(\hat{\boldsymbol{\theta}})$ denotes the open ball of radius a centered at x) and let $L := N(a, \mathring{S}^a)$ denote the covering number of $\mathring{S}^a$. Note that $L \leq N(a, S^a)$. Let $B_1, \dots, B_L$ denote closed balls of radius a whose union contains $\mathring{S}^a$. Let $E_1, \dots, E_L$ be defined as $E_1 := B_1$ and $E_i := B_i \setminus (\cup_{j<i} B_j)$ for $i = 2, \dots, L$. We can also ensure that $\cup_{i=1}^L E_i = \mathring{S}^a$ by removing the set $\mathring{S}^a \setminus \cup_i E_i$ from each set E_i.

Suppose that a probability measure H is chosen so that G and H satisfy (49) on each set E_i for $i = 1, \dots, L$. So, after setting $c = 3$ and $m = \lfloor 13.5\bar{n}a^2/\underline{s} \rfloor$, we obtain

$$\max_{1\leq k\leq K} \left| f_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - f_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \right| \leq \frac{2}{(2\pi\underline{s})^{d/2}} \left\{ \left(\frac{9e\bar{n}a^2}{2\underline{s}(m+1)} \right)^{m+1} + e^{-\underline{n}a^2/(2\bar{s})} \right\}$$

and

$$\max_{1\leq k\leq K} \left\| \mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - \mathbf{q}_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \right\| \leq \frac{6\sqrt{\bar{n}}a}{(2\pi\underline{s})^{d/2}} \left\{ \left(\frac{9e\bar{n}a^2}{2\underline{s}(m+1)} \right)^{m+1} + \sqrt{\frac{\bar{s}}{e\bar{n}a^2}} + e^{-\underline{n}a^2/(2\bar{s})} \right\}.$$

Since $m + 1 \geq 13.5\bar{n}a^2/\underline{s}$,

$$\left\{ \frac{9e\bar{n}a^2}{2\underline{s}(m+1)} \right\}^{m+1} \leq \left(\frac{e}{3} \right)^{m+1} \leq e^{-(m+1)/12} \leq \exp\left(-\frac{13.5\bar{n}a^2}{12\underline{s}} \right) \leq e^{-\bar{n}a^2/(2\underline{s})}.$$

Then, the result for the first inequality follows. This also gives

$$\begin{aligned} \max_{1\leq k\leq K} \left\| \mathbf{q}_{k,G}(\underline{\hat{\boldsymbol{\theta}}}_k) - \mathbf{q}_{k,H}(\underline{\hat{\boldsymbol{\theta}}}_k) \right\| &\leq \frac{6\sqrt{\bar{n}}a}{(2\pi\underline{s})^{d/2}} \left\{ e^{-\bar{n}a^2/(2\underline{s})} + \sqrt{\frac{\bar{s}}{e\bar{n}a^2}} + e^{-\underline{n}a^2/(2\bar{s})} \right\} \\ &\leq \frac{6a}{(2\pi\underline{s})^{d/2}} \left\{ \sqrt{\frac{\bar{s}}{ea^2}} + 2\sqrt{\bar{n}}e^{-\underline{n}a^2/(2\bar{s})} \right\}. \end{aligned}$$

$\square$

Finally, we control the covering number of S^a under a pseudo metric.

Lemma E.3. *Assume that $\bar{n}/\underline{n}$ tends to some positive constant as $\underline{n} \to \infty$. There exists a positive constant $\underline{n}_0 > 0$ such that for any $\underline{n} > \underline{n}_0$ and every compact set $S \subset \mathbb{R}^d$, $M > 0$ and $\eta \le \underline{s}^{-d/2} \wedge e^{-1}$, we have*

$$\log N(\eta, \mathbb{F}, \|\cdot\|_{\infty,S^a}) \vee N(\eta, \mathbb{F}_q, \|\cdot\|_{\infty,S^a}) \le_d N(a, S^a) \log^{d+1} \frac{c_{d,\underline{s},\bar{s},\bar{s}'}}{\eta} \tag{50}$$

for $a = \sqrt{\frac{2\bar{s}}{\underline{n}} \log \frac{1}{\eta}}$.

Proof. Let $G \in \mathcal{P}(\mathbb{R}^d)$, and obtain a discrete distribution H supported on S^a with at most l as in Lemma E.2.

First inequality. Let $\mathcal{C}$ denote a minimal ζ-net of S^a, and let H' approximate each atom of H with its closest element from $\mathcal{C}$. Writing $H = \sum_{j=1}^l w_j \delta_{\mathbf{a}_j}$ and $H' = \sum_{j=1}^l w_j \delta_{\mathbf{b}_j}$, we have

$$\begin{aligned}
\|f_{\bullet,H} - f_{\bullet,H'}\|_{\infty,S^a} &= \max_{k\in[K]} \sup_{\mathbf{x}\in S^a} |f_{k,H}(\mathbf{x}) - f_{k,H'}(\mathbf{x})| \\
&\le \max_{k\in[K]} \sup_{\mathbf{x}\in S^a} \sum_j w_j \left|\varphi^{(k)}(\mathbf{x}; \mathbf{a}_j) - \varphi^{(k)}(\mathbf{x}; \mathbf{b}_j)\right| \\
&\le_d \frac{\sqrt{\bar{n}} + 2d^{3/2}\bar{s}'}{\underline{s}^{(d+1)/2}} \zeta.
\end{aligned}$$

In the third line, we used Lemma D.2 and $\|\mathbf{a}_j - \mathbf{b}_j\| \le \zeta$. Let $\mathcal{D}$ denote a minimal ξ-net of $\Delta^{l-1} = \{\mathbf{w} \in \mathbb{R}_+^l : \sum_{j=1}^l w_j = 1\}$ in the ℓ_1 norm, and approximate the weights w by their closest element $v \in \mathcal{D}$. Writing $H'' = \sum_{j=1}^l v_j \delta_{\mathbf{b}_j}$,

$$\begin{aligned}
\|f_{\bullet,H'} - f_{\bullet,H''}\|_{\infty,S^a} &= \max_{k\in[K]} \sup_{\mathbf{x}\in S^a} |f_{k,H'}(\mathbf{x}) - f_{k,H''}(\mathbf{x})| \\
&\le \max_{k\in[K]} \sup_{\mathbf{x}\in S^a} \sum_j |w_j - v_j| |\varphi^{(k)}(\mathbf{x}; \mathbf{b}_j)| \le_d \underline{s}^{-d/2} \xi.
\end{aligned}$$

Applying the triangle inequality to the past three displays,

$$\|f_{\bullet,G} - f_{\bullet,H''}\|_{\infty,S^a} \le_{d,\underline{s},\bar{s}'} e^{-\underline{n}a^2/(2\bar{s})} + (\sqrt{\bar{n}} + 1)\zeta + \xi.$$

Letting $\xi := \eta$, $\zeta := \frac{\xi}{\sqrt{\bar{n}}+1}$, and selecting $a = \sqrt{\frac{2\bar{s}}{\underline{n}} \log \frac{1}{\xi}}$ yields the convergence rate $\|f_{\bullet,G} - f_{\bullet,H''}\|_{\infty,S^a} \le C_{d,\underline{s},\bar{s}'}\eta$. To take a as such, we let $\eta < \frac{1}{e}$ so that $\log(1/\xi) > 1$.

The number of possible H'' is

$$|\mathcal{C}|\cdot|\mathcal{D}|= N(\xi,\Delta^{l-1})\binom{N(\zeta,S^a)}{l} \le \left[\left(1+\frac{2}{\xi}\right)\frac{eN(\zeta,S^a)}{l}\right]^l$$

by Stirling's lower bound and standard arguments. From the previous Lemma,

$$l=\left(\left\lfloor\frac{27\bar{n}a^2}{\underline{s}}\right\rfloor+2\right)^d N(a,S^a)+1>\left(\frac{54\tau\bar{n}}{\underline{n}}+1\right)^d N(a,S^a)+1\ge_d\left(\frac{\tau\bar{n}}{\underline{n}}\right)^d N(a,S^a)$$

with a,ξ,η selected as above, so using the arguments in Saha and Guntuboyina [2020],

$$\frac{N(\zeta,S^a)}{l}\le\left\{\frac{\underline{n}}{\tau\bar{n}}\left(1+\frac{a}{\zeta}\right)\right\}^d\le\left(\frac{\underline{n}}{\bar{n}}+2\sqrt{\frac{\bar{s}\underline{n}}{\bar{n}\eta^3}}\right)^d\le\left(1+2\sqrt{\frac{\bar{s}}{\eta^3}}\right)^d.$$

So, as $1/\eta>e$,

$$\begin{aligned}\log N(C_{d,\underline{s},\bar{s}'}\eta,\mathbb{F},\|\cdot\|_{\infty,S})&=\log\left[\left(1+\frac{2}{\xi}\right)\frac{eN(\zeta,S^a)}{l}\right]^l\le l\left\{\log\frac{3}{C_{d,\underline{s},\bar{s}'}\eta}\frac{eN(\zeta,S^a)}{l}\right\}\\&\le_d N(a,S^a)\left\{\frac{\bar{n}}{\underline{n}}\log(1/\eta)\right\}^d\log\left\{\frac{3}{C_{d,\underline{s},\bar{s}'}\eta}\left(1+2\sqrt{\frac{1}{C_{d,\underline{s},\bar{s}'}\eta^3}}\right)^d\right\}\\&\le N(a,S^a)\left(\frac{\bar{n}}{\underline{n}}\right)^d\log^d\frac{1}{\eta}\log\frac{c_{d,\underline{s},\bar{s}'}}{\eta^{1+3d/2}}\\&\le_d N(a,S^a)\log^{d+1}\left(\frac{c_{d,\underline{s},\bar{s}'}}{\eta}\right).\end{aligned}$$

Second inequality. We use the same covering argument to prove the second inequality as well. For $\mathcal{C}$, a minimal ζ-net of S^a and H', the approximate of H with closest element from $\mathcal{C}$, $H=\sum_{j=1}^l w_j\delta_{\mathbf{a}_j}$ and $H'=\sum_{j=1}^l w_j\delta_{\mathbf{b}_j}$, we have

$$\begin{aligned}\|\mathbf{q}_{\bullet,H}-\mathbf{q}_{\bullet,H'}\|_{\infty,S^a}&\le\max_{k\in[K]}\sup_{\mathbf{x}\in S^a}\sum_j w_j\|(\sqrt{n_k}\mathbf{a}_j-\mathbf{x})\varphi^{(k)}(\mathbf{x};\mathbf{a}_j)-(\sqrt{n_k}\mathbf{b}_j-\mathbf{x})\varphi^{(k)}(\mathbf{x};\mathbf{b}_j)\|\\&\le\max_{k\in[K]}\sup_{\mathbf{x}\in S^a,\mathbf{u}_j\in I(\mathbf{a}_j,\mathbf{b}_j)}\sum_j w_j\|\sqrt{n_k}(\mathbf{a}_j-\mathbf{b}_j)\varphi^{(k)}(\mathbf{x};\mathbf{a}_j)+(\sqrt{n_k}\mathbf{b}_j-\mathbf{x})\langle\mathbf{b}_j-\mathbf{a}_j,\nabla_{\mathbf{u}_j}\varphi^{(k)}(\mathbf{x};\mathbf{u}_j)\rangle\|\\&\le\zeta\max_{k\in[K]}\left\{\frac{\sqrt{n_k}}{\underline{s}^{d/2}}+\sup_{\mathbf{x}\in S^a}\sup_{\mathbf{u}_j\in I(\mathbf{a}_j,\mathbf{b}_j)}(\|\mathbf{x}-\sqrt{n_k}\mathbf{u}_j\|+\sqrt{n_k}\|\mathbf{u}_j-\mathbf{b}_j\|)\cdot\|\nabla_{\mathbf{u}_j}\varphi^{(k)}(\mathbf{x};\mathbf{u}_j)\|\right\}\\&\le\zeta\max_{k\in[K]}\left\{\frac{\sqrt{n_k}}{\underline{s}^{d/2}}+\frac{\sqrt{n_k}\zeta(\sqrt{n_k}+2d^{1.5}\bar{s}')}{\underline{s}^{(d+1)/2}}+\sup_{\mathbf{x}\in S^a}\sup_{\mathbf{u}_j\in I(\mathbf{a}_j,\mathbf{b}_j)}\|\mathbf{x}-\sqrt{n_k}\mathbf{u}_j\|\cdot\|\nabla_{\mathbf{u}_j}\varphi^{(k)}(\mathbf{x};\mathbf{u}_j)\|\right\}\end{aligned}$$

where $I(\mathbf{a}_j, \mathbf{b}_j)$ denotes a set of linear interpolations between $\mathbf{a}_j, \mathbf{b}_j$. In the third line, we used (35) and Lemma D.2. Plus, for $\mathbf{z}_{\mathbf{u}_j} := \Sigma_k(\mathbf{u}_j)^{-1/2}(\mathbf{x} - \sqrt{n_k}\mathbf{u}_j)$, we have

$$\begin{aligned}
\sup_{\mathbf{x}\in S^a} \sup_{\mathbf{u}_j \in I(\mathbf{a}_j,\mathbf{b}_j)} \|\Sigma_k(\mathbf{u}_j)^{1/2}\mathbf{z}_{\mathbf{u}_j}\| \cdot \left\| \nabla_{\mathbf{u}_j} \frac{\phi_d(\mathbf{z}_{\mathbf{u}_j})}{|\Sigma_k(\mathbf{u}_j)|^{1/2}} \right\| & \\
\leq \frac{\bar{s}^{1/2}}{\underline{s}^{(d+1)/2}} \sup_{u\geq 0} u \left\{ u\left(\sqrt{\bar{n}} + \bar{s}'u\right) + d^{1.5}\bar{s}' \right\} \phi_1(u) & \\
\leq \frac{2\bar{s}^{1/2}}{\sqrt{e}\underline{s}^{(d+1)/2}} \left(\sqrt{\bar{n}} + \bar{s}'d^{1.5}\right). &
\end{aligned}$$

So, we obtain

$$\|\mathbf{q}_{\bullet,H} - \mathbf{q}_{\bullet,H'}\|_{\infty,S^a} \leq \frac{\sqrt{\bar{n}}\zeta}{\underline{s}^{d/2}} \left\{ 1 + \frac{(\zeta + 2\bar{s}^{1/2})(1 + 2d^{1.5}\bar{s}'/\sqrt{\bar{n}})}{\underline{s}^{1/2}} \right\}.$$

Let $\mathcal{D}$ denote a minimal ξ-net of $\Delta^{l-1} = \{\mathbf{w} \in \mathbb{R}_+^l : \sum_{j=1}^l w_j = 1\}$ in the ℓ_1 norm, and approximate the weights w by their closest element $v \in \mathcal{D}$. Writing $H'' = \sum_{j=1}^l v_j \delta_{\mathbf{b}_j}$,

$$\begin{aligned}
\|\mathbf{q}_{\bullet,H'} - \mathbf{q}_{\bullet,H''}\|_{\infty,S^a} &= \max_{k\in[K]} \sup_{\mathbf{x}\in S^a} \sum_j (w_j - v_j) \|(\sqrt{n_k}\mathbf{b}_j - \mathbf{x})\varphi^{(k)}(\mathbf{x};\mathbf{b}_j)\| \\
&\leq \sum_j |w_j - v_j| \frac{\bar{s}^{1/2}}{\underline{s}^{d/2}} \sup_{u\geq 0} u\phi_1(u) \lesssim_d \bar{s}^{1/2}\underline{s}^{-d/2}\xi.
\end{aligned}$$

Finally, aggregating the inequalities for large enough $\bar{n}$,

$$\|\mathbf{q}_{\bullet,G} - \mathbf{q}_{\bullet,H''}\|_{\infty,S^a} \lesssim_{d,\underline{s},\bar{s},\bar{s}'} \sqrt{\bar{n}} \left[ae^{-\underline{n}a^2/(2\bar{s})} + \frac{1+\xi}{\sqrt{\bar{n}}} + \zeta(\zeta+1) \right].$$

Now we set $\xi = \eta$, $\zeta = \eta/\sqrt{\bar{n}}$, and $\eta < 1/e$. Then, $a = \sqrt{\frac{2\bar{s}}{\underline{n}} \log \frac{1}{\xi}}$ yields the convergence rate $\|\mathbf{q}_{\bullet,G} - \mathbf{q}_{\bullet,H''}\|_{\infty,S^a} \leq C_{d,\underline{s},\bar{s},\bar{s}'}\sqrt{\bar{n}}(a + 1/\sqrt{\bar{n}})\eta =: \tilde{\eta}$ (so $\tilde{\eta} \lesssim_{\bar{s}} \sqrt{\bar{n}}a\eta$).

Next, we had

$$|\mathcal{C}|\cdot|\mathcal{D}| \leq \left[\left(1 + \frac{2}{\xi}\right) \frac{eN(\zeta, S^a)}{l} \right]^l, \quad l \gtrsim_d \left(\frac{\tau\bar{n}}{\underline{n}}\right)^d N(a, S^a)$$

and

$$\frac{N(\zeta, S^a)}{l} \leq \left\{ \frac{\underline{n}}{\tau\bar{n}} \left(1 + \frac{a}{\zeta}\right) \right\}^d \leq \left(1 + \sqrt{\frac{2\underline{n}\bar{s}}{\bar{n}\xi^3}}\right)^d \underset{\underline{n}\to\infty}{\leq} \left(\sqrt{\frac{C_{d,\bar{s}}}{\eta^3}}\right)^d.$$

So, as $1/\eta > e$,

$$\begin{aligned}\log N(\tilde{\eta}, \mathbb{F}_q, \|\cdot\|_{\infty,S}) &\leq l \left\{ \log \frac{3}{\xi} \frac{eN(\zeta, S^a)}{l} \right\} \\ &\leq_d N(a, S^a) \left\{ \log(1/\xi) \right\}^d \log \left(\frac{C'_{d,\bar{s}}}{\eta^{1+3d/2}} \right) \leq_d N(a, S^a) \log^{d+1} \frac{C'_{d,\bar{s}}}{\eta}\end{aligned}$$

hence

$$\log N(\eta, \mathbb{F}_q, \|\cdot\|_{\infty,S}) \leq_d N(a, S^a) \log^{d+1} \frac{\sqrt{\bar{n}} a C'_{d,\underline{s},\bar{s},\bar{s}'}}{\eta} \leq N(a, S^a) \log^{d+1} \frac{C'_{d,\underline{s},\bar{s},\bar{s}'}}{\eta}.$$

$\square$